VisionIQ - Google Cloud Structure

Consolidated reference and current-status supplement | 2 November 2026

Project: gbr-mts-wgh-hkt-dev **Region:** europe-west3 (Frankfurt)

Document intent. One cumulative, portable reference preserving the 30 October FND-03 master guide (51 pages) and the 31 October FND-04 deployment update (14 pages), preceded by this November 2 current-state supplement. Older original pages are retained unmodified as dated evidence, not silently rewritten as current configuration.

Current infrastructure and application status

Component	Status at latest supplied evidence
Cloud SQL / VPC	Cloud SQL visioniqdev, DB visioniq_dev, private IP 10.102.16.8; VPC connector vpc-con; migration job previously succeeded.
Artifact Registry	team-visioniq-docker-dev; three release images pushed 31 Oct with reviewed Git SHA tag 02779fd00c0c0800443486a7263c7ee19d6f8fc0.
Cloud Run services	visioniq-api: READY revision visioniq-api-00001-9gm, port 5100; visioniq-intelligence: READY revision visioniq-intelligence-00001-zfn, port 8000. Both internal/authenticated; each 100% traffic on 31 Oct.
Cloud Run migration job	visioniq-db-migrate previously executed successfully; FND-04 did not execute or update it.
Cloud Storage	visioniq-source-evidence-dev used for canonical LRN-01 archival tests; 5/5 sampled source archive hashes matched.
Cloud Scheduler / ingestor job	No verified scheduled LRN-01 Cloud Run Job or Cloud Scheduler trigger. Continuous cloud LEARN not yet established.
Cloud Build / Developer Connect	Automation deferred / blocked by corporate connection authorization; manual deployment is the documented path.

Verification boundary: Cloud Run revisions are deployed and metadata/startup-verified, but authenticated in-network /health and /ready, real service-to-Cloud-SQL queries and role/secret-isolation checks were deferred. Local LRN-01 tests do not prove a cloud-hosted ingestion worker.

Contents and source precedence

Part A (this supplement): Current architecture, local ingestion evidence, open gates, safe operations and change log.

Part B: Original 30 October Cloud Structure master (51 pages; page labels preserved; FND-03 provisioning, SQL, network, registry, IAM, Cloud Run Jobs, screenshots and history).

Part C: Original 31 October FND-04 Deployment Update (14 pages; exact deployment evidence, 3-image release procedure, runtime status, troubleshooting, commands and pending acceptance).

Precedence: For conflicts, use the most recent dated evidence (this supplement, then 31 Oct FND-04, then 30 Oct FND-03). Historic descriptions such as "application services not deployed" and "no evidence bucket verified" remain factual only at the old snapshot date. Do not treat any as live Google Cloud inventory.

1. Architecture - deployed versus planned

Present cloud infrastructure: Artifact Registry images are deployed as separate internal/authenticated Cloud Run API and Intelligence services. Both are configured for private network access to Cloud SQL through the existing VPC connector. Whether their actual application-level SQL operations work is not yet proven by available HTTP/database tests.

Current LRN-01 evidence path: Windows workstation -> .NET ingestor CLI -> Azure DevOps REST -> local isolated PostgreSQL + real GCS evidence bucket. The PostgreSQL test endpoint was 127.0.0.1:15543, not Cloud SQL. The canonical archive was tested against GCS, but this does not imply LRN-01 is deployed as a cloud job.

Planned continuous-learning path: approved Cloud Scheduler / event trigger -> dedicated ingestor Cloud Run Job -> ADO REST -> managed Cloud SQL checkpoints and canonical records -> GCS snapshots -> downstream LEARN processing. This path needs explicit authorization, packaging, runtime IAM, networking, verification and scheduling.

System flow (text diagram)

ADO sources (work items, PRs, commits, changed files, builds) → LRN-01 .NET ingestion → canonical archive (GCS) + evidence/checkpoints (PostgreSQL) → LRN-03 to LRN-14 downstream processing. A separate FND-04 path runs the .NET API and Python Intelligence Cloud Run HTTP services.

2. LRN-01: real-source test evidence - 2 November 2026

Test	Observed outcome
Historical full work-item export	367 items, 1,002 links, 0 not stored/quarantined; local JSONL run exit 0.
Work-item comments	IDs 106675 / 107328 / 115804: expected 9 / 2 / 0; observed 9 / 2 / 0, no gap/truncation.
Git history	KIOSK_TVM initial source sync: 64 commits and 12 PRs; later local dated commit export: 39 commits.
Changed files	Selected real KIOSK_TVM commit 6f13cdd9f97eb57ecb0c60f9bc40724cee8bb6a9: 8 VisionIQ files vs 8 ADO files, 0 differences in path/change type, not truncated.
CI builds	Canonical run 00c39f65-5491-4b63-b78b-6ff5176d1f49: 86 builds, 149 links, 112 build-link graph assertions, 0 failures/quarantines, exit 0.
Canonical persistence	Isolated PostgreSQL checkpoints verified across work-items, commits and PRs; GCS archive existence sampled and 5/5 SHA-256 matches.
Failure handling	30/30 targeted failure/retry tests passed, 0 skipped.
Full solution regression	402 tests passed, 0 failed/skipped, exit 0, build succeeded with 66 warnings.

Scope is the specifically approved Genoa Kiosk project with Git repository KIOSK_TVM. The build stream is project-scoped; ADO_REPOS=KIOSK_TVM does not alone prove pipeline-filter enforcement. The tested changed-file match is a sample, not proof of exhaustive repository fidelity.

Source-of-truth feature docs in GitHub: docs/features/LRN-01-history-connectors/ and docs/LRN-01-Test-Report.md on feature/learn. PR #14 remains DRAFT.

3. Cloud Run deployment and operational boundary

Service	31 Oct verified state / remaining validation
visioniq-api	READY revision visioniq-api-00001-9gm; internal ingress + authentication; port 5100; startup observed. Authenticated internal HTTP and SQL role checks pending.
visionIQ-intelligence	Actual Cloud Run resource name visioniq-intelligence; READY revision visioniq-intelligence-00001-zfn; port 8000; internal/authenticated; startup observed. In-network HTTP and SQL checks pending.
visioniq-db-migrate	Separate one-shot Cloud Run JOB. Previous successful migration job validated its own private SQL connectivity only; not an application readiness test.

Security and configuration requirements

- Cloud Run services use shared runtime Google identity visioniq-runtime in FND-04; explicit application secret grants are recorded, but shared-identity acceptance is not formal security approval.
- Application DB roles visioniq_api and visioniq_intelligence are separate; actual deployed-runtime SQL role permissions and cross-schema write denial require authorized tests.
- No public ingress/unauthenticated access should be enabled to make laptop Postman tests pass. Prior laptop proxy requests received Google Frontend HTTP 404, not a validated application /health failure.
- Secret Manager passwords, Azure DevOps PATs, secret payloads, personal labels and unredacted IAM principal lists must not be placed in the master guide or repository.

4. Practical read-only cloud inventory

Approved Windows PowerShell session; these commands inspect resources only. They do not confirm HTTP readiness or live ADO source access.

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION = 'europe-west3'
gcloud config get-value project
gcloud run services list --project=$PROJECT --region=$REGION
gcloud run jobs list --project=$PROJECT --region=$REGION
gcloud artifacts docker images list europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev --project=$PROJECT
--include-tags
gcloud sql instances describe visioniqdev --project=$PROJECT
gcloud compute networks vpc-access connectors describe vpc-con --project=$PROJECT --region=$REGION
gcloud builds connections list --project=$PROJECT --region=$REGION
gcloud builds triggers list --project=$PROJECT --region=$REGION
```

5. Unresolved checks and handoff

- FND-04: approved in-network /health and /ready for both deployed services; verify an actual Cloud SQL read through each runtime DB role; verify database and secret isolation, not just resource readiness.
- LRN-01 cloud rollout: dedicated ingestor image, Cloud Run Job, runtime permissions, Cloud SQL connectivity, Scheduler/event trigger, monitoring and overlap protection not yet deployed or evidenced.
- FND-AUTO: approved corporate GitHub / Developer Connect connection and Cloud Build release automation remain deferred.
- Review 66 .NET build warnings for severity; LRN-01 functional criteria have supporting evidence but formal administrative acceptance remains separate.

- Before demo, perform an authorized, controlled backfill 3-4 days before demo, validate downstream knowledge, then rely on delta sync and a tested snapshot. Do not blindly repeat --full on shared immutable evidence.

6. Change log (this supplement)

2026-10-30: Master FND-03 infrastructure inventory; migration job operational, API and intelligence services not yet deployed at that date.

2026-10-31: Manual three-image release to Artifact Registry; API and Intelligence Cloud Run revisions READY; runtime integration acceptance deferred.

2026-11-02: LRN-01 live work-item/comments/commit/changed-file/build tests and isolated canonical storage/GCS hash checks completed; 402/402 full regression tests pass. No dedicated cloud ingestion job or scheduler asserted.

Evidence retention: Append sanitized cloud screenshots and exact execution metadata to the corresponding service sections after any future approved deployment; do not overwrite incident history.

VisionIQ — Google Cloud Structure

Permanent Cloud Infrastructure & Operations Reference
Google Cloud project: `gbr-mts-wgh-hkt-dev`
Primary region: `europe-west3` (Frankfurt)
Last reviewed: 30 October 2026 · FND-03 validated; FND-04 source prepared, Cloud Build connection attempts failed, deployment access pending. **FND-04 user stories remain OPEN.**

This document describes the **Google Cloud infrastructure used by VisionIQ**, organized by **cloud service** rather than implementation phase. It explains the architecture, purpose, configurations, resource relationships, operations, known issues and verification evidence. Future infrastructure must be recorded by extending the relevant existing chapter or adding a new cloud-service chapter **in this same document**.

Document boundaries: This guide covers Google Cloud resources and infrastructure operations. For first-time developer installation, use **Complete VisionIQ Setup**. For everyday workstation startup and shutdown, use **Run VisionIQ Locally**. Historical milestones and task identifiers, where useful, appear only in the implementation history and change log.

Evidence and safety: This is a dated infrastructure snapshot, not a live Google Cloud query. Where historical creation commands or logs were not preserved, the text says so. Commands that create, update, execute or delete resources must never be run solely to reproduce documentation. Do not store passwords, private keys, tokens, personally identifying labels or unredacted secret outputs.

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1. Cloud Environment Overview

Infrastructure at a glance

```
Google Cloud project: gbr-mts-wgh-hkt-dev  (europe-west3)

 Artifact Registry (team-visioniq-docker-dev)
    └── Migration image: visioniq-intelligence:fnd03-netretry
          └── Cloud Run JOB: visioniq-db-migrate
                ├── Service account: visioniq-db-migrate
                ├── Secret Manager: 3 database-password secrets
                └── Serverless VPC Access connector: vpc-con
                      └── Corporate VPC / connector subnet
                            └── Private Service Access (PSA)
                                  └── Cloud SQL: visioniqdev
                                        └── Database: visioniq_dev
                                              ├── workflow schema
                                              ├── evidence schema
                                              └── knowledge schema

 Cloud Run application SERVICES: .NET API / Python Intelligence — NOT DEPLOYED
 Cloud Scheduler / Cloud Storage / Pub/Sub app integrations — NOT CONFIGURED HERE
```

Verified network route: Both the saved Cloud Run job YAML and the **successful execution's own Networking tab** identify Serverless VPC Access connector `vpc-con` with private-IP-only traffic routing. Direct VPC egress was discussed earlier but is **not** the verified successful route.

Resource inventory

Google Cloud component	Actual name / configuration	Status and purpose
Cloud SQL	`visioniqdev`, PostgreSQL 16.15; `visioniq_dev` database	Created; migration completed and database checked
Artifact Registry	`team-visioniq-docker-dev`	Created; hosts the image used by the migration job
Secret Manager	Three database password secrets	Created; migration job references all three; migration and shared runtime identities both have direct Secret Accessor grants on the two application-password secrets (30 Oct)
Service accounts	`visioniq-db-migrate`; `visioniq-runtime`	Both created; migration identity used by job; runtime identity now granted access to two application secrets, but application Cloud Run services are not deployed
Cloud Run Jobs	`visioniq-db-migrate`	Created; one task completed successfully
VPC + PSA	Existing corporate network	Reused, not necessarily created by VisionIQ
Serverless VPC Access	`vpc-con`	Verified on successful execution; creation provenance not yet confirmed

Google Cloud component	Actual name / configuration	Status and purpose
Cloud Run Services	.NET API and Python Intelligence	Not deployed; deployment pipeline source prepared for corporate approval
Cloud Scheduler	No confirmed VisionIQ schedule	Not configured
Cloud Build source connection	Proposed `visioniq-build-dev`	Two creation attempts failed; no successful connection shown
GitHub authorization secret	`visioniq-build-token-dev`	User reports creation and secret-level Cloud Build accessor grant; version/ token scopes not independently verified
FND-04 user stories	Not a cloud resource	**All OPEN / deferred**, pending approved source integration and actual deployment

Scope of verification: A completed migration job establishes database migration connectivity and checks, not application deployment or functioning external engineering-data integrations.

How to extend this document

When a resource is added or changed, update its **service chapter** with the purpose, project, location, creation method, exact settings, dependencies, IAM/secret references, commands or Console navigation, verification logs, screenshot evidence, problems and corrections, security/cost impact, and reversal/ recovery considerations. Update **Section 17: Infrastructure Change Log**. Do not create a new master cloud guide for each feature or foundation milestone.

2. Google Cloud Project and Environments

Primary environment

Setting	Recorded value
Project	`gbr-mts-wgh-hkt-dev`
Region	`europe-west3` (Frankfurt, Germany)
Cloud SQL instance connection name	`gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`
Workload model today	One-shot Cloud Run migration job; future app services not deployed
Local environment	Independent Docker PostgreSQL; no automatic connection to Cloud SQL

Project authentication and read-only verification (PowerShell)

```
gcloud auth list
gcloud config get-value project
gcloud projects list --format="table(projectId,name)"
gcloud config list
```

Use an approved company identity and verify the selected project **before** any state-changing command. A selected project is not proof of sufficient IAM authorization. Corporate project IAM updates were previously permission-denied and requested through cloud administrators.

Verified environment and inventory snapshot — 30 October 2026

The following observations were obtained with **read-only Google Cloud CLI commands** from a Windows PowerShell terminal. They are a dated snapshot of the shared development project, **not a live query** and **not proof of VisionIQ deployment**. The authenticated active account was an approved **corporate identity**; personal account addresses are deliberately omitted.

Inspection	Verified output	Meaning / qualification
`gcloud auth list`	Corporate identity active; other signed-in identities omitted	CLI authentication confirmed. Authentication is not proof of authorization for deployment or IAM updates.
`gcloud config get-value project`	`gbr-mts-wgh-hkt-dev`	Correct active project selected.
`gcloud run services list --region=$REGION`	Five existing Cloud Run services were listed; **none were named `visioniq-api` or `visioniq-intelligence`**	Other applications share this project. Do **not** change or mislabel them as VisionIQ. This list is region-scoped.
`gcloud artifacts repositories list --location=$REGION`	Six repositories listed; VisionIQ repository `team-visioniq-docker-dev` exists	Other five repositories belong to other project workloads or Cloud Functions; they are not VisionIQ resources by assumption.
`gcloud iam service-accounts list --filter="email:visioniq"`	`visioniq-runtime` and `visioniq-db-migrate`; both `DISABLED=False`	Both Google service account identities exist and are enabled. Their detailed IAM bindings were **not** checked.
`gcloud builds triggers list --region=$REGION`	`Listed 0 items`	No Cloud Build triggers visible **in `europe-west3`** for the caller. Does not prove no triggers exist in other locations or external CI systems.
`gcloud compute networks vpc-access connectors describe vpc-con`	`state: READY`, VPC `vpc-mts-wgh-hkt-mtswghhkt-dev`, connector subnet `subnet-mts-wgh-hkt-mtswghhkt-dev-connector`	Connector exists and is ready. Runtime service-to-Cloud-SQL connectivity still requires an independent application test.

Command syntax correction: `gcloud run services list` accepts `--region=$REGION` for this inspection; `--LOCATION=$REGION` produced `unrecognized arguments`. The corrected command completed successfully. Do not reproduce the failing command in routine runbooks.

Configuration and resource labeling

The Cloud Run job has resource labels including `hackathon_team=vision_iq` and an owner label, the value of which is intentionally omitted from the shared handbook. For future resources, prefer consistent non-personal labels such as application, environment and component where organizational policy permits. Labels are for classification and billing analysis; they are **not IAM controls**.

3. Cloud SQL and PostgreSQL

Instance identity and capacity

Setting	Recorded configuration
Project	`gbr-mts-wgh-hkt-dev`
Instance	`visioniqdev`
Instance connection name	`gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`
Database engine	PostgreSQL 16.15 (Console snapshot 29 Oct 2026)
Edition	Enterprise
Region / primary zone	`europe-west3` (Frankfurt) / `europe-west3-c`
Availability	Single zone
Machine	`db-custom-2-4096` (2 vCPUs, 4 GB RAM)
Data cache	Disabled
Storage	10 GB SSD; automatic storage increase enabled
Encryption	Google-managed key
Password policy	Enabled
Application database	`visioniq_dev` (verified 30 Oct 2026)
Database administrator username	`visionIQ` (**case-sensitive PostgreSQL identifier**)
Creation method	Manual Google Cloud Console provisioning; **not Terraform-managed**
Resource labels	Two labels were visible in Console; their personal/team identifier values are omitted

Database security and certificates

Setting	Recorded configuration
TLS	Allow only SSL connections enabled
Trusted client certificates	Not required
Server CA	Google-managed CAS
Server certificate expiry	29 Oct 2027 at 5:44:49 PM as shown; time zone not stated
Automatic server certificate rotation	Disabled (review with cloud administrator)
Cloud SQL flag	`cloudsql.iam_authentication=on`
Google Cloud services authorization	Disabled
Data API access	Disabled

Setting	Recorded configuration
App Engine authorization	Enabled
Migration connection authentication	PostgreSQL username/password secrets; IAM database authentication was **not** used by this migration runner
Migration TLS client configuration	`sslmode=require` (encrypts, but is not the same as hostname/certificate verification via `verify-full`)

Backups, recovery, maintenance and database insights

Setting	Recorded configuration
Backup tier	Standard
Automated backups	Enabled; 7 backups retained
Backup window	12:30–16:30 IST, as shown in Console
PITR	Enabled; logs retained 7 days
Backup location	EU multi-region (`eu`)
Deletion protection	Enabled
Retain backups after deletion	Enabled
Final backup on deletion	Enabled; retention 60 days
Maintenance timing	Week 1; update window may be any day
Next maintenance	None scheduled in 29 Oct snapshot
Maintenance version	`POSTGRES_16_15.R20260712.01_44`
Query Insights	Enabled
Store client IP / application tags	Both disabled
Maximum recorded query length	1024
Sampling rate	5 as displayed; unit not specified in supplied screenshot
Console toggles	Vertex AI integration enabled; Knowledge Catalog integration enabled. Neither proves VisionIQ application's AI integration is functional.

Cost and data-residency review: The single-zone instance, storage, backups, PITR, and Frankfurt/EU placement require normal organizational approval; no current billing figure or formal data-residency approval was provided.

Database users and schema permissions

PostgreSQL role	Purpose	Privileges validated by migration `verify`
`"visionIQ"`	Cloud SQL admin/migration account	Executes migration DDL, role creation/ bootstrap and verification; case-sensitive quoted spelling matters in manual SQL.
`visioniq_api`	.NET API DB login	`SELECT` in three schemas; INSERT/ UPDATE/DELETE in `workflow`, `evidence`; cannot write `knowledge` or create schema objects.
`visioniq_intelligence`	Python Intelligence DB login	`SELECT` in three schemas; INSERT/ UPDATE/DELETE in `knowledge`; cannot write `workflow` / `evidence` or create schema objects.

The three database passwords must be **distinct**. Runtime roles are not PostgreSQL superusers and do not receive DDL rights. Default table/sequence grants are configured for future objects created by the migration administrator. The admin owns and applies schema migrations; a runtime 'writer' is a grantee, not necessarily the schema owner.

Alembic migration and database checks

The migration uses an ordered Alembic revision history. Its recorded first revision is `20261029_0001`; the verified schemas are `workflow`, `evidence`, `knowledge`, with `vector` extension and full-text testing. Database role bootstrap and write-isolation checks were run from the Cloud Run migration job. SQL Studio provided an additional verification of schema, extension and Alembic head. The cloud migration method and commands are in Sections 8 and 13.

```
SELECT version_num FROM public.alembic_version;
SELECT schema_name FROM information_schema.schemata
 WHERE schema_name IN ('workflow','evidence','knowledge') ORDER BY schema_name;
SELECT extname FROM pg_extension WHERE extname = 'vector';
SELECT to_tsvector('english','historical defect regression')
  @@ plainto_tsquery('english','defect') AS fulltext_ok;
```

These are read-only validation queries. They do not on their own prove restricted-role write denials; that requires the database migration verifier with the correct identities.

4. VPC, Private Networking and Connectivity

Existing network and private access

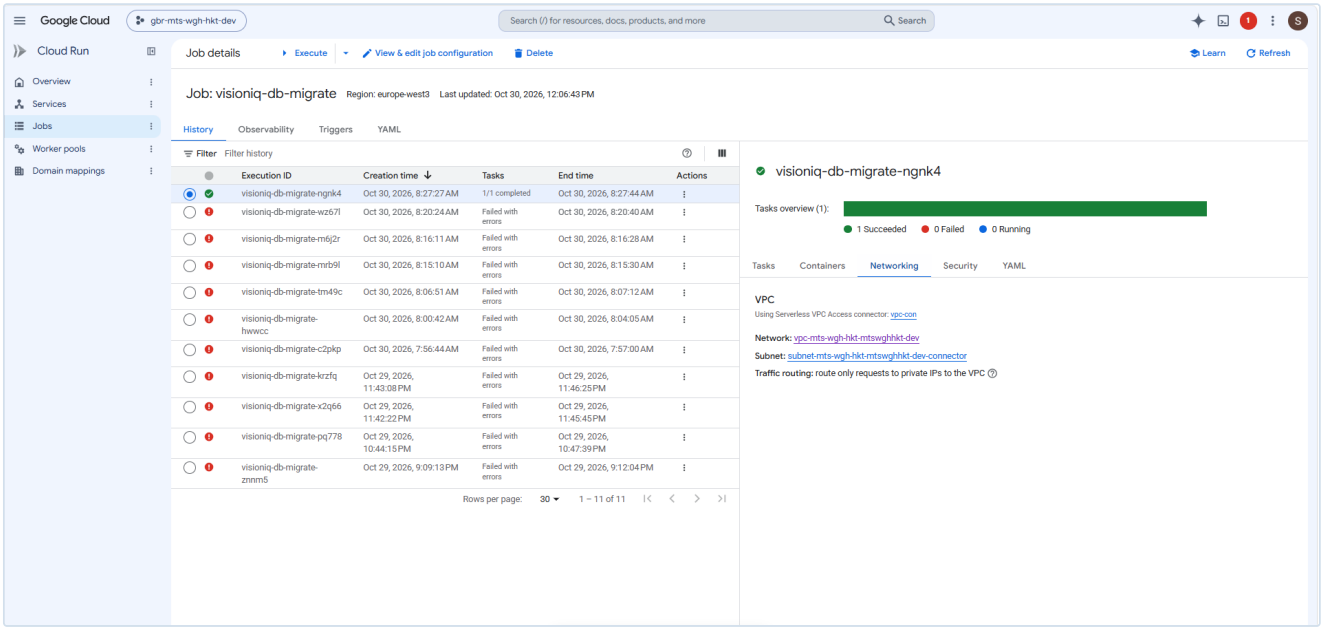
Setting	Recorded configuration / evidence
VPC network	`vpc-mts-wgh-hkt-mtswghhkt-dev`
Cloud SQL private connection	Private Service Access (PSA) enabled
Cloud SQL private IPv4	`10.102.16.8`
Cloud SQL port	`5432`

Setting	Recorded configuration / evidence
Public IP	Disabled
Allocated private range	`10.102.16.0/20` (service-networking address allocation)
Allocated address name	`services-subnet-mts-wgh-hkt-dev`
Observed peerings	`servicenetworking-googleapis-com` ACTIVE; `peering-to-shared-services` ACTIVE (30 Oct investigation)
Observed firewall	`postgres-egres`, priority 1000, enabled, destination `10.102.16.0/20` (rule details need admin confirmation)
Cloud Run migration-job egress (successful execution AND current saved job)	**Serverless VPC Access connector** `vpc-con`, `private-ranges-only` in `europe-west3`. **VERIFIED** by execution `visioniq-db-migrate-ngnk4` Networking tab; also confirmed by saved job generation 10 YAML.
Connector `vpc-con` and subnet	**VERIFIED in successful 08:27 execution**: network `vpc-mts-wgh-hkt-mtswghhkt-dev`, subnet `subnet-mts-wgh-hkt-mtswghhkt-dev-connector`. Egress routes **only requests to private IPs** through the VPC.
Laptop-to-private-IP route	Not verified; corporate workstation/Zscaler is not automatically on the VPC
Permission constraint	`gcloud services vpc-peerings list` previously failed with `servicenetworking.services.get` denied; the Compute network peering listing succeeded
Shared-network approval	Reuse of existing VPC has been submitted to the cloud team for confirmation; do not change company network/firewall settings without approval

Security boundary: The migration Cloud Run Job demonstrated reachability to Cloud SQL through the private VPC. It does not establish that the future .NET/Python Cloud Run services have working VPC routes; configure and validate each service in FND-04.

Verified connection used by Cloud Run migration

The successful Cloud Run execution `visioniq-db-migrate-ngnk4` used **Serverless VPC Access connector** **`vpc-con`**. Its execution-specific Networking tab shows the corporate VPC, the connector subnet and routing of private-IP requests. The current saved job YAML independently names the same connector. The network and PSA were reused; the connector creation command and original ownership remain undocumented.



Evidence: Google Cloud Console > Cloud Run > Jobs > `visioniq-db-migrate` > History > select execution `visioniq-db-migrate-ngnk4` > **Networking**. Screenshot provided 30 October 2026. Screenshot is from the successful *execution itself* rather than the job editor.

Execution Networking field	Observed value	Evidence level
Execution status	Green succeeded; `1/1` task completed (consistent with earlier Tasks screenshot)	**VERIFIED**
Egress mechanism	**Using Serverless VPC Access connector**	**VERIFIED**
Connector name	`vpc-con`	**VERIFIED**
VPC network	`vpc-mts-wgh-hkt-mtswghhkt-dev`	**VERIFIED**
Connector subnet	`subnet-mts-wgh-hkt-mtswghhkt-dev-connector`	**VERIFIED**
Traffic routing	**Route only requests to private IPs to the VPC**	**VERIFIED**
Cloud SQL destination	`visioniqdev` private IP `10.102.16.8:5432` (from Cloud SQL configuration and migration history)	**Corroborated by separate evidence**

Reconciled finding: The successful initial migration's actual network path was **Cloud Run migration job -> Serverless VPC Access connector `vpc-con` -> VPC `vpc-mts-wgh-hkt-mtswghhkt-dev` -> PSA -> Cloud SQL private IP**. Earlier notes referencing **Direct VPC egress** must be retained only as *historical investigation*, not as the successful final route. The screenshot is sufficient to resolve the final execution's VPC mode even though the job YAML's last update followed that execution. Do not infer why the earlier attempts failed without their logs.

Documentation consequence: Treat initial migration connectivity as verified for the migration job only. **Future .NET and Python Cloud Run services must separately configure and test VPC connectivity**; migration-job networking does not automatically apply to other services.

Connector capacity and readiness snapshot — 30 October 2026

Read-only connector inspection returned:

Property	Observed value
State	`READY`
Network	`vpc-mts-wgh-hkt-mtswghhkt-dev`
Subnet	`subnet-mts-wgh-hkt-mtswghhkt-dev-connector`
Region	`europe-west3`
Machine type	`e2-micro`
Minimum instances	`2`
Maximum instances	`3`
Reported minimum throughput	`200` (as displayed by the CLI)
Reported maximum throughput	`300` (as displayed by the CLI)

This is connector **configuration and readiness**, not an end-to-end connectivity test of future Cloud Run application services. The previous migration-job success remains the separate application-to-database proof for that job only. Do not recreate or resize a shared connector without the relevant infrastructure approvals.

Earlier alternative investigated

Direct VPC egress was discussed/configured during investigation, but that is not the network route demonstrated by the successful execution. Retain it in the troubleshooting history, rather than presenting Direct VPC as the final architecture. A local laptop or Cloud SQL Auth Proxy does not automatically establish a private route into the corporate VPC.

Read-only VPC inventory commands

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION = 'europe-west3'
gcloud compute networks peerings list --network=vpc-mts-wgh-hkt-mtswghhkt-dev --project=$PROJECT
gcloud compute addresses list --global --filter='purpose=VPC_PEERING' --project=$PROJECT
gcloud compute firewall-rules list --project=$PROJECT --filter='network:vpc-mts-wgh-hkt-mtswghhkt-dev AND
direction:EGRESS'
gcloud compute networks vpc-access connectors describe vpc-con --region=$REGION --project=$PROJECT
```

A previous Service Networking inspection failed with `servicenetworking.services.get` permission denied. This does not mean no peering exists; Compute network peerings were observed separately. Do not modify shared network/firewall rules without approved access.

5. Artifact Registry

Repository and deployed image

Configuration	Recorded value
Artifact Registry Docker repository	`team-visioniq-docker-dev`
Location	`europe-west3`
Registry hostname	`europe-west3-docker.pkg.dev`
Image path used by migration job	`europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev/visioniq-intelligence:fnd03-netretry`
Status	Repository created; job references the image; successful run recorded
Later image packaging	A separate `db/Dockerfile` was introduced in source, but not confirmed deployed to the job

Artifact Registry stores deployable images, **not a running Cloud Run service**. The saved image tag is not an immutable digest. Capture the digest and associated source commit during future releases.

Registry inventory verification — 30 October 2026

The `gcloud artifacts repositories list --project=$PROJECT --location=$REGION` command returned the VisionIQ Docker repository `team-visioniq-docker-dev`, location `europe-west3`, format `DOCKER`, mode `STANDARD_REPOSITORY`, Google-managed encryption. The CLI snapshot reported approximately **220.899 MB** of repository storage. This value is a snapshot, not a budget or a live cost metric. The existing repository's labels included a team identifier and a personally identifying owner label; individual owner values are intentionally excluded from this shared guide.

Five additional Artifact Registry repositories were visible in the same project. They are shared-project resources and **not VisionIQ-owned by inference**. This inspection does not confirm which Docker image tags or digests are currently stored; use the read-only image inventory commands below to verify those separately.

Inspect images without changing Cloud resources

```
gcloud artifacts repositories list --project=gbr-mts-wgh-hkt-dev --location=europe-west3
gcloud artifacts docker images list europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev --
project=gbr-mts-wgh-hkt-dev --include-tags
```

Creation, image build and publishing record

The precise historic repository creation commands and image push transcript are not present in the evidence currently retained. A reference workflow is recorded in Appendix A, clearly marked as a **state-changing future operation**, not a claim about which command was executed in October. Maintain this chapter when publishing additional .NET/Python images and record image digest, revision, build source, push log and artifact retention policy.

Evidence to attach

Artifact Registry → Repositories → `team-visioniq-docker-dev` (repository metadata), plus Docker Images (tag, digest, upload date). Do not show unrelated corporate repositories.

6. Secret Manager

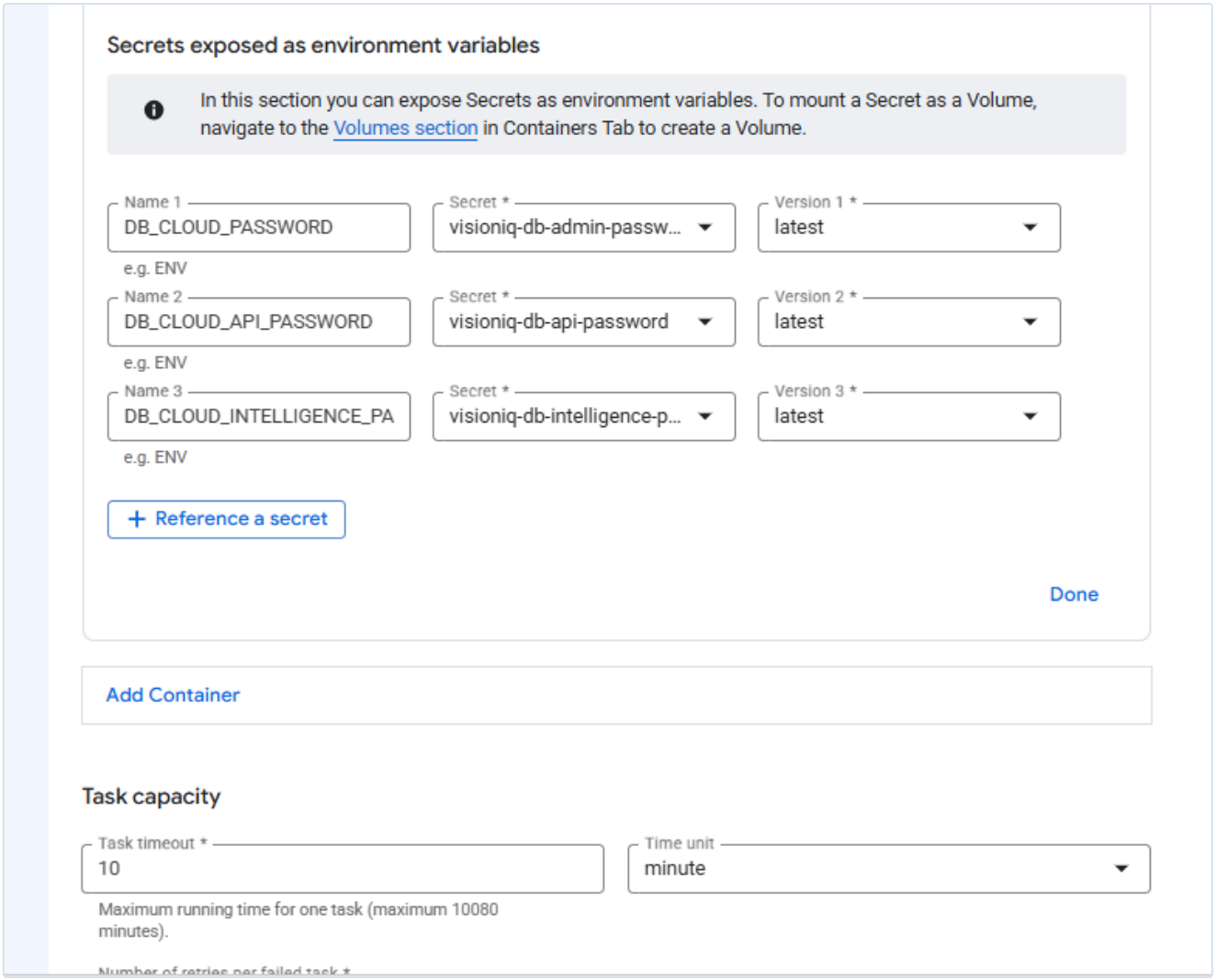
Configured database credentials

Environment variable on migration job	Secret name
`DB_CLOUD_PASSWORD`	`visioniq-db-admin-password:latest`
`DB_CLOUD_API_PASSWORD`	`visioniq-db-api-password:latest`
`DB_CLOUD_INTELLIGENCE_PASSWORD`	`visioniq-db-intelligence-password:latest`

Secret values, password policy responses, access tokens and service-account JSON keys **must never** be copied into this guide, console screenshots shared publicly, tickets or Git. Direct IAM grants on the two **application-password** secrets were inspected on 30 October 2026, as recorded below. The administrator-password secret IAM policy was **not** included in that inspection.

Job secret bindings

The Cloud Run editor and saved YAML show three Secret Manager references with version selector `latest`. Their values are not displayed or reproduced. The migration image consumed the credentials to authenticate as the database administrator and test restricted API/Intelligence roles.



Current security limitation: Using `latest` simplifies development but reduces reproducibility for historical executions when versions rotate; future releases should explicitly document rotation and access strategy. Record secret *names, versions and IAM bindings*, never contents.

Application-secret IAM verification and access update — 30 October 2026

Before the runtime grants, a read-only `gcloud secrets list --format="value(name)"` showed all three VisionIQ database-password secrets in the project. The output also contained other projects' Git/ADO/ Confluence credentials and pre-existing Cloud Build/GitLab integration secret names; these **belong to the shared project**, are not listed here, and do **not** establish that VisionIQ can use an active Cloud Build repository connection. **No secret value or token was opened or retrieved.**

Secret (before IAM update)	Direct Secret Accessor binding returned	Runtime identity explicitly listed?	Historical observation
`visioniq-db-api-password`	`serviceAccount:visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com` — `roles/secretmanager.secretAccessor`	**No**	Direct secret-level migration grant VERIFIED; runtime direct grant NOT SHOWN

Secret (before IAM update)	Direct Secret Accessor binding returned	Runtime identity explicitly listed?	Historical observation
`visioniq-db-intelligence-password`	`serviceAccount:visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com` — `roles/secretmanager.secretAccessor`	**No**	Direct secret-level migration grant VERIFIED; runtime direct grant NOT SHOWN
`visioniq-db-admin-password`	**Not inspected in this verification**	Unknown	Used by migration job; do not make it available to application runtime accounts

Historical interpretation: The initial read-only inspection above listed only direct bindings. It did **not** establish that the runtime identity was effectively denied access, because inherited project/folder permissions were not evaluated. **Those earlier results have been superseded for the two application secrets by the subsequent explicit grants below.** No IAM change was made during the *initial inspection*.

Verified IAM changes — shared runtime identity (later on 30 October 2026)

The approved CLI session executed `gcloud secrets add-iam-policy-binding` on each **application-password** secret. Both commands returned `Updated IAM policy`, with the two principals below appearing together under `roles/secretmanager.secretAccessor`:

Secret	Existing direct accessor retained	Direct accessor added	Result
`visioniq-db-api-password`	`visioniq-db-migrate@...`	`visioniq-runtime@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	**GRANTED — binding confirmed in CLI response**
`visioniq-db-intelligence-password`	`visioniq-db-migrate@...`	`visioniq-runtime@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	**GRANTED — binding confirmed in CLI response**

Only these **two secret-level IAM policies** were changed. The administrator password (`visioniq-db-admin-password`) was **not** part of these updates; no new runtime access to it was shown. No project-wide Secret Accessor role was granted by these commands. The CLI output confirms the policy updates, **not** an actual Cloud Run startup or secret-read attempt.

Security design decision implemented: Both future application services are intended to use the **same `visioniq-runtime` Google identity**. That identity now has direct access to *both* application passwords. This is the earlier hackathon trade-off, **not** independent per-service secret isolation. The distinct PostgreSQL `visioniq_api` and `visioniq_intelligence` roles still limit writes when each application uses its own credentials. Corporate acceptance of this compromise against the original independent-identity requirement remains to be confirmed before final milestone closure.

Next security decision: The shared runtime account is now actually granted access to both application secrets. If the original FND-04 acceptance criterion of independent identities must be satisfied without an approved exception, create distinct .NET and Python runtime service accounts, each with access to its own secret, and remove the shared identity from those secrets after successful cutover. The two application Cloud Run services have not yet been deployed.

Cloud Build GitHub authorization secret — 30 October 2026 (later verification)

New resource reported by operator: `visioniq-build-token-dev`, within project `gbr-mts-wgh-hkt-dev`. The operator confirmed creating this secret through the Secret Manager Console and assigning the Google-managed Cloud Build service agent **Secret Manager Secret Accessor on the secret**. This is a GitHub **authentication-token secret**, **not** a database password and not a Cloud Run application runtime secret. The actual secret version, replication setting, token type/scopes, GitHub App installation authorization and effective access were not independently verified from Console output. No value is recorded.

Field	Status / evidence
Secret resource	`visioniq-build-token-dev` — creation user-confirmed in Console
Intended contents	GitHub authorization token for a manual Cloud Build connection; never print or retrieve for documentation
Intended accessor	`service-494928274920@gcp-sa-cloudbuild.iam.gserviceaccount.com` — secret-level accessor grant user-confirmed
Used by an active VisionIQ Cloud Build connection?	**No confirmed connection; both attempts failed**
Cleanup / risk	**Unused PAT should be revoked at its GitHub source and the unneeded secret retired through approved change control if the approach is abandoned.** Check project policy and audit logging without copying the token

Safety: Do not confuse this secret with `visioniq-db-api-password`, `visioniq-db-intelligence-password`, or `visioniq-db-admin-password`. Do not put access tokens in scripts, logs, tickets, chats, screenshots or the cloud guide. Because corporate GitHub App installation permission was unavailable, the intended connection was deferred; storing a PAT does not eliminate GitHub organization approval requirements.

Evidence and maintenance

Secret Manager → Secrets list (names only); secret metadata, replication/location and service-account secret-access bindings from approved read-only views. Do not open or copy a secret payload for documentation. The cloud migration job and future application services should have access only to required secrets.

7. IAM and Service Accounts

Created Google service accounts

Principal	Status / intended usage
`visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	**EXISTS; used by successful Cloud Run migration job.** Restrict access to migration/bootstrap database secrets.
`visioniq-runtime@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	**EXISTS; direct `roles/secretmanager.secretAccessor` grants VERIFIED on the two app-password secrets (30 Oct).** Intended for both services; Cloud Run usage, broader effective access and deployer IAM remain pending.

Google IAM roles and PostgreSQL database roles are **not the same**. Assigning a Google service account to Cloud Run does not automatically grant table access within PostgreSQL. The application requires its own restricted database login/password supplied securely.

Pending IAM request, sent 30 Oct: Project IAM role updates were denied to the requesting corporate identity. Cloud administrators were asked for appropriate IAM role configuration, Cloud Scheduler access and approval to reuse the VPC. `roles/resourcemanager.projectIamAdmin` and `roles/iam.serviceAccountAdmin` were discussed/requested, but are **not granted or confirmed**. Prefer admin-managed grants rather than broad standing IAM admin access.

Current and intended access boundaries: the migration account has direct access to both application-password secrets for bootstrap/verification; its administrator-secret policy was not inspected in the latest IAM check. The shared runtime identity has now been directly granted access to both application-password secrets. Because the future .NET and Python Cloud Run services are intended to use the same Google identity for hackathon simplicity, both could resolve either secret; keeping distinct PostgreSQL logins limits writes only if each application uses its intended credentials. Actual service deployment, any inherited permissions, and Vertex AI access are not yet verified.

Service account existence check — 30 October 2026

The filtered CLI list confirmed the following **enabled** Google service accounts. This check establishes existence and disabled state only; it does not establish effective Secret Manager, Artifact Registry, Cloud Run invocation, Cloud Build, or deployment permissions.

Service account	DISABLED	Present usage
`visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	`False`	Verified as the Cloud Run migration job's configured service account
`visioniq-runtime@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	`False`	Application-secret-level grants later verified; active Cloud Run service binding and deployer IAM not yet verified

IAM and Cloud Build readiness verification — 30 October 2026

The approved corporate Google Cloud CLI identity ran **read-only** pre-deployment checks against `gbr-mts-wgh-hkt-dev`. No resource was created, no IAM binding was changed, and no service was deployed. User account email addresses and personal labels are not recorded in this document.

Read-only check	Observed result	Accurate interpretation / next action
`gcloud services list --enabled` filtered to five APIs	`artifactregistry.googleapis.com`, `cloudbuild.googleapis.com`, `iam.googleapis.com`, `run.googleapis.com`, `secretmanager.googleapis.com` all present	These core APIs are enabled. **This does not imply deployment authorization.**
`gcloud builds connections list --region=europe-west3`	`Listed 0 items` (confirmed twice)	No visible **second-generation Cloud Build host connections** in this region. No approved corporate Git integration established by these results; check provider/connection policy before creating any.

Read-only check	Observed result	Accurate interpretation / next action
`gcloud projects get-iam-policy` with member filter for `visioniq-runtime`	No rows displayed	No matching **explicit project-level allow-policy binding was returned** for that principal. This does not exclude inherited IAM roles or per-secret/per-resource permissions.
`gcloud iam service-accounts get-iam-policy visioniq-runtime@...`	`etag: ACAB`, no `bindings`	No **direct allow-policy bindings** returned on the service account resource. This policy primarily governs who can act on that account; it is **not** a summary of permissions the account has elsewhere. Inherited permissions are not shown.

Security interpretation: A Cloud Run *deployer* normally needs `roles/iam.serviceAccountUser` on the service identity along with Cloud Run deployment permissions; a Cloud Run *runtime identity* needs `roles/secretmanager.secretAccessor` on each approved secret that it consumes. Grant only the appropriate **runtime** database secret(s), never the migration administrator secret to the application service. For the originally proposed shared `visioniq-runtime` identity, giving access to both application secrets creates a common secret-access boundary even though PostgreSQL roles remain separate; this is a documented hackathon compromise, not strict per-service isolation. If the original FND-04 independent-identity gate must be satisfied without exception, provision separate .NET/Python runtime identities after administrator approval.

Current deployment gate: `PENDING` — the **two direct secret-level grants are now VERIFIED** by successful policy-update responses. Effective ability to resolve secret references at Cloud Run startup, deployer permissions, inherited IAM and pipeline access are **not yet verified**. Cloud Build is enabled but the repository connection and pipeline remain unconfigured in the inspected region. The corporate GitHub Enterprise hosted-runner restriction is separate from Cloud Build repository-connection authorization.

Workload identity versus PostgreSQL access

Google Cloud service accounts control workload access to Google resources; PostgreSQL roles control access **inside** Cloud SQL. Using the migration service account does not itself grant database table privileges. The successful job used a service account with Secret Manager-backed database passwords. Runtime service-account creation does **not** mean runtime Cloud Run services are deployed or permissions are complete.

Read-only service-account inventory

```
gcloud iam service-accounts list --project=gbr-mts-wgh-hkt-dev --format='table(email,displayName,disabled)'
gcloud run jobs describe visioniq-db-migrate --project=gbr-mts-wgh-hkt-dev --region=europe-west3
```

Record requested/approved IAM roles and their effective principal scopes after the cloud administrator responds; avoid assuming broad project IAM administrator roles were granted.

Cloud Build service agent — IAM findings and effective-access investigation (30 October 2026)

Cloud Build's failed connection operation explicitly identified **Google-managed principal** `service-494928274920@gcp-sa-cloudbuild.iam.gserviceaccount.com`, not the human operator and not `visioniq-runtime`. The role listing for this principal returned the following **direct project-level allow bindings**:

Project-level role	Observed in CLI output
`roles/cloudbuild.connectionAdmin`	Yes
`roles/cloudbuild.serviceAgent`	Yes
`roles/iam.serviceAccountUser`	Yes
`roles/secretmanager.admin`	Yes
`roles/secretmanager.secretAccessor`	Yes

Policy Analyzer screenshot (operator-provided): indicated that `roles/secretmanager.admin` contains/assigns `secretmanager.secrets.create` for the principal. This is a role-grant analysis, **not proof of effective authorization**. **Policy Troubleshooter screenshot (operator-provided):** overall result **UNKNOWN**; deny-policy evaluation reported **PERMITS access**, allow-policy evaluation was **UNKNOWN / conditionally granted**, and Principal Access Boundary evaluation was **UNKNOWN due to missing visibility**. It would be incorrect to call this a confirmed deny-policy violation or a confirmed service-agent misconfiguration. An authorized organization IAM administrator needs to inspect the actual binding conditions, inherited policies, Principal Access Boundary and Cloud Audit Logs.

Both connection attempts returned **permission denied** for `secretmanager.secrets.create` despite the reported bindings. Do not add more project-wide roles simply to retry. In the shared corporate project, `roles/secretmanager.admin` and other broad roles must be reviewed by the administrator, including removing temporary grants that are no longer needed. These project-level roles are **not** the design goal for a dedicated VisionIQ pipeline build identity or test runner.

8. Cloud Run Jobs

Purpose and actual deployed configuration

Setting	Recorded value
Job name	`visioniq-db-migrate`
Region / project	`europe-west3` / `gbr-mts-wgh-hkt-dev`
Service account	`visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`
Cloud SQL target	`gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`
Logical DB / host / port	`visioniq_dev` / `10.102.16.8` / 5432
Last **recorded** deployed image	`europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev/visioniq-intelligence:fnd03-netretry`
New source design	Separate image built with `db/Dockerfile`; **not yet confirmed deployed to Cloud Run**
Job command	`python -m db.cloud_manage all --instance gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`
CPU / memory	1 vCPU / 1 GiB (job description recorded 29 Oct)
Tasks / parallelism	1 / 1

Setting	Recorded value
Timeout / retries	10 min / 0 retries
Egress	**VERIFIED for successful 08:27 execution AND current saved job:** Serverless VPC Access connector `vpc-con` ; private IPs only. Earlier notes about Direct VPC are historical troubleshooting.
Authentication	PostgreSQL password secrets, TLS `sslmode=require`

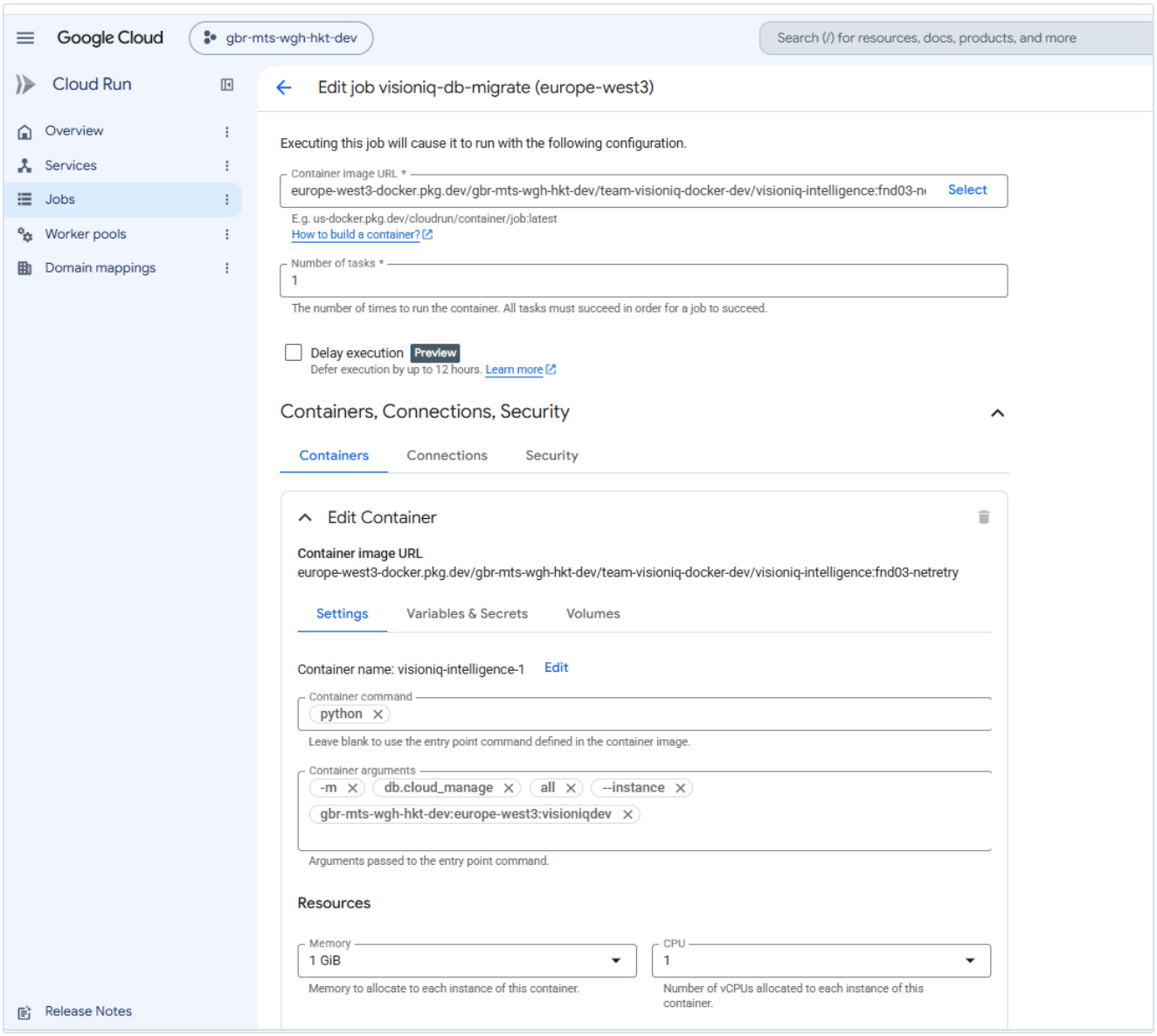
Recorded non-secret job variables: `DB_TARGET=cloud`, `DB_CLOUD_INSTANCE=gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`, `GCP_PROJECT_ID=gbr-mts-wgh-hkt-dev`, `DB_CLOUD_NAME=visioniq_dev`, `DB_CLOUD_USER=visionIQ`, `DB_HOST=10.102.16.8`, `DB_PORT=5432`, `DB_API_USER=visioniq_api`, `DB_INTELLIGENCE_USER=visioniq_intelligence`, `VISIONIQ_CLOUD_RUN_MIGRATIONS=approved`.

Important distinction: The previously deployed image successfully ran the migration. Later source changes introduced a dedicated migration image; do **not** claim the Cloud Run job has been redeployed with the new image. Before a future cloud migration, build/tag/publish an approved artifact from `db/Dockerfile`, update the job image, inspect the configured job, then execute under the normal deployment approval procedure. Avoid rerunning an admin migration job merely to synchronize the image tag with local development.

Console screenshots — container, variables, security and saved YAML

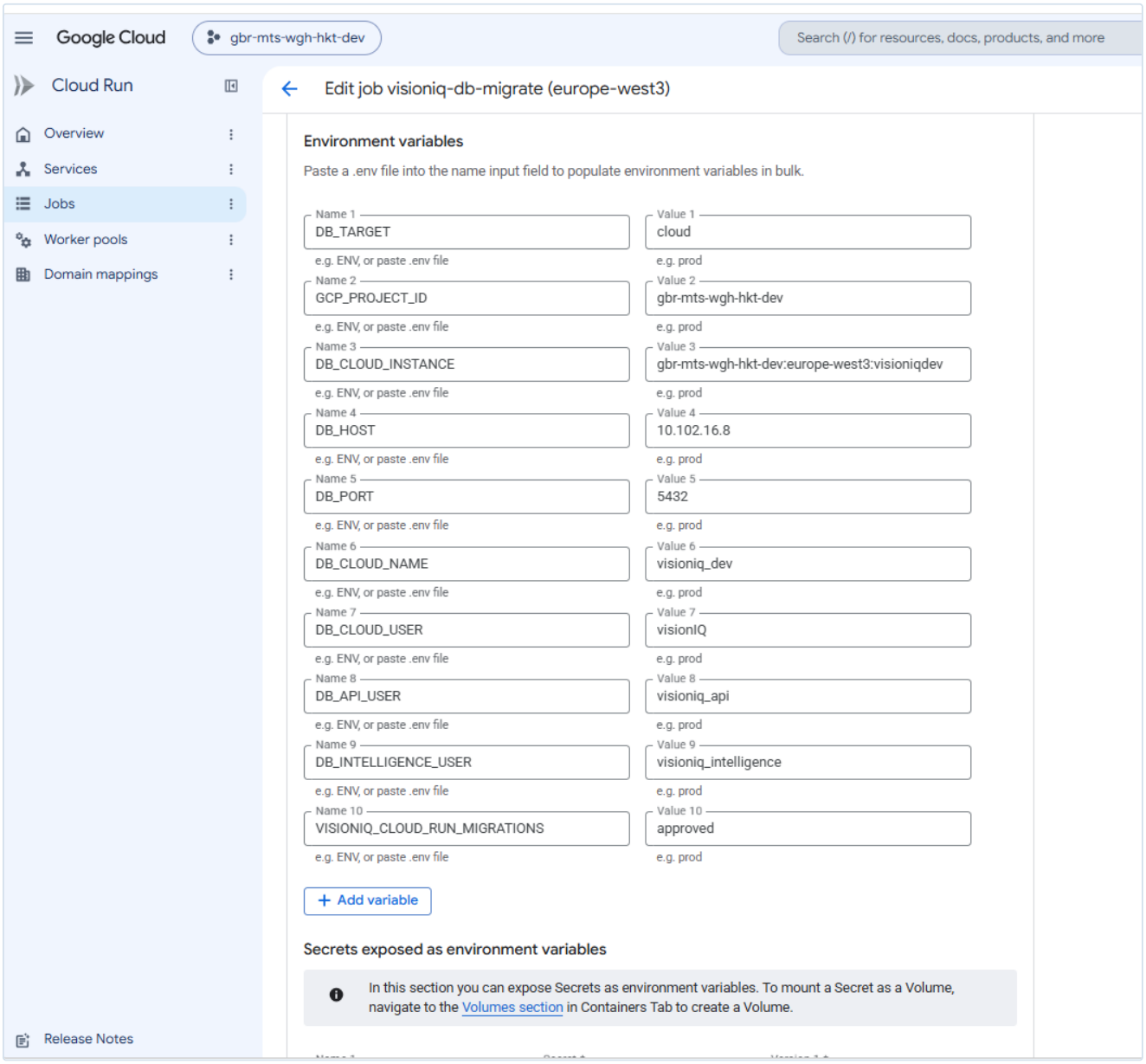
Evidence source: Five earlier screenshots supplied on 30 October 2026 from **Google Cloud Console -> Cloud Run -> Jobs -> Edit job** `visioniq-db-migrate` **(** `europe-west3` **)**. They are reproduced below in readable crops; the package includes the full originals. These are **editor views**, not execution logs, an update confirmation, or a deployment diff. Do not click **Update** or **Execute job immediately** just to document them.

CR-01 - Container image, entry command, resources



Observed: Project `gbr-mts-wgh-hkt-dev`; job `visioniq-db-migrate`; container image `europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev/visioniq-intelligence:fnd03-netretry`; **one task**; container named `visioniq-intelligence-1`; command `python`; arguments `-m db.cloud_manage all --instance gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`; **1 GiB RAM, 1 vCPU**. This is the earlier intelligence-based migration image; the newer dedicated `db/Dockerfile` image is not evidenced as deployed.

CR-02 - Explicit non-secret environment variables



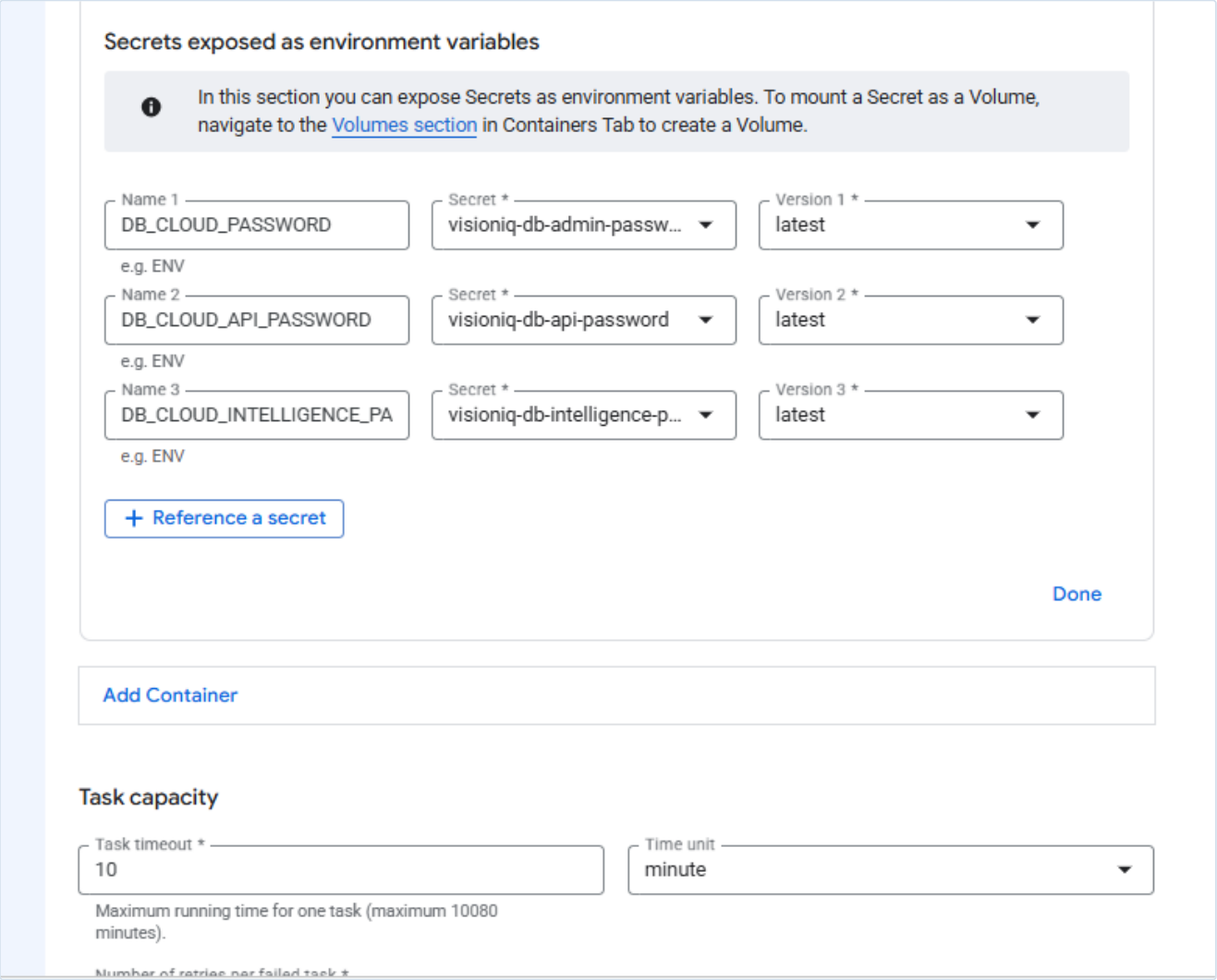
Observed keys and values:

Key	Value shown in editor
`DB_TARGET`	`cloud`
`GCP_PROJECT_ID`	`gbr-mts-wgh-hkt-dev`
`DB_CLOUD_INSTANCE`	`gbr-mts-wgh-hkt-dev:europe-west3:visioniqdev`
`DB_HOST`	`10.102.16.8`
`DB_PORT`	`5432`
`DB_CLOUD_NAME`	`visioniq_dev`
`DB_CLOUD_USER`	`visionIQ` (**case-sensitive**)
`DB_API_USER`	`visioniq_api`

Key	Value shown in editor
`DB_INTELLIGENCE_USER`	`visioniq_intelligence`
`VISIONIQ_CLOUD_RUN_MIGRATIONS`	`approved`

These values configure the migration runner. Their presence does not prove PostgreSQL login succeeded or that the cloud job was executed using this unsaved editor state; those conclusions require execution logs.

CR-03 - Secret Manager references, not plaintext credentials

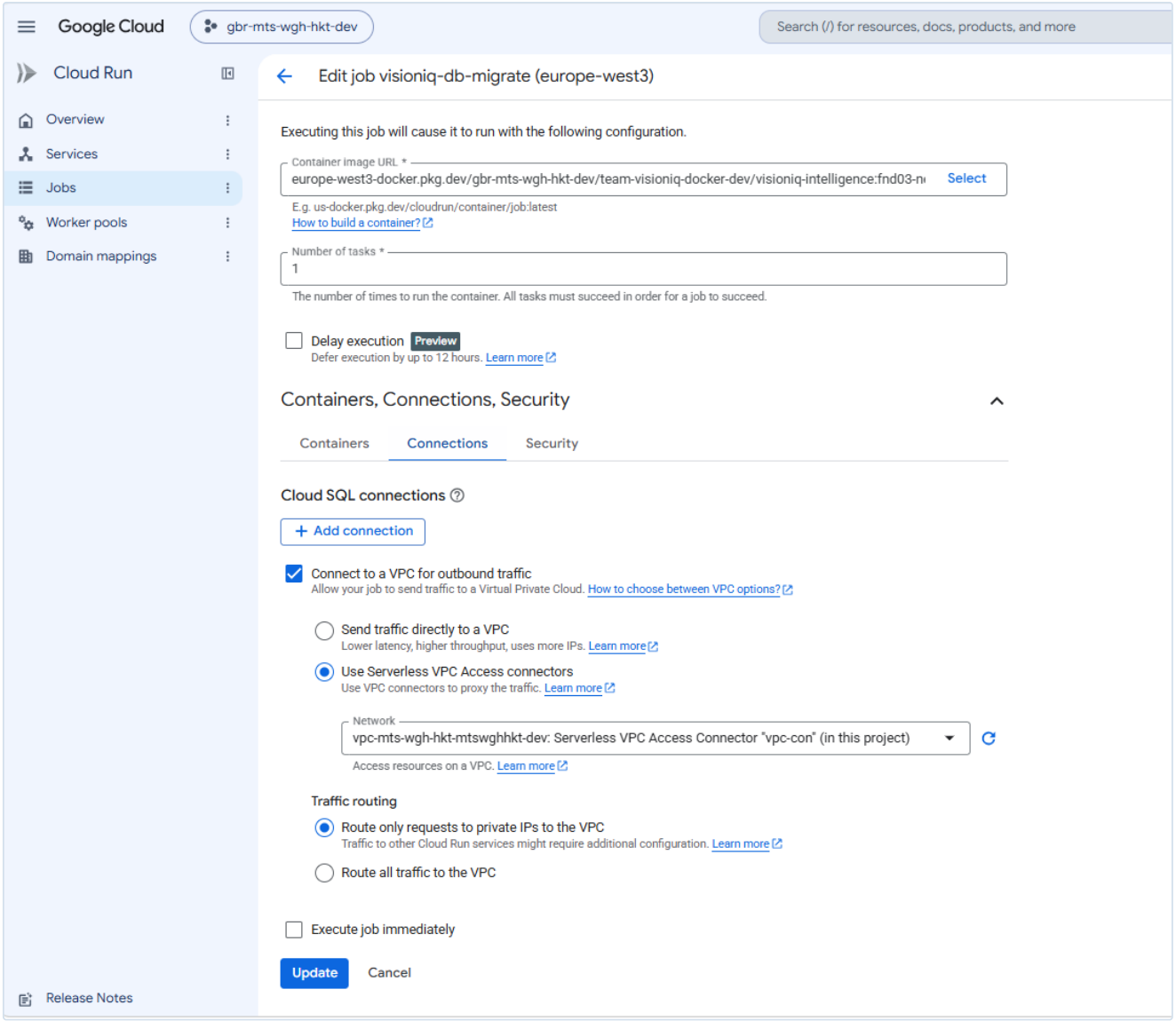


Observed: Three secrets are referenced as environment variables and select version `latest`. The screenshot visually truncates portions of the first and third secret names and the final environment variable name; the **full names below are also supported by the earlier implementation records** (not independently legible end-to-end in this screenshot).

Environment key (historical runbook)	Secret name (historical runbook)	Version visible
`DB_CLOUD_PASSWORD`	`visioniq-db-admin-password`	`latest`
`DB_CLOUD_API_PASSWORD`	`visioniq-db-api-password`	`latest`
`DB_CLOUD_INTELLIGENCE_PASSWORD`	`visioniq-db-intelligence-password`	`latest`

The screenshot does **not** show secret values. Do not add screenshots of the Secret Manager plaintext **Versions -> View secret value** screen to any shared guide. `latest` is mutable; deployments requiring deterministic recovery should record approved version metadata without copying the values.

CR-04 - Job editor networking (verified against execution CR-08)

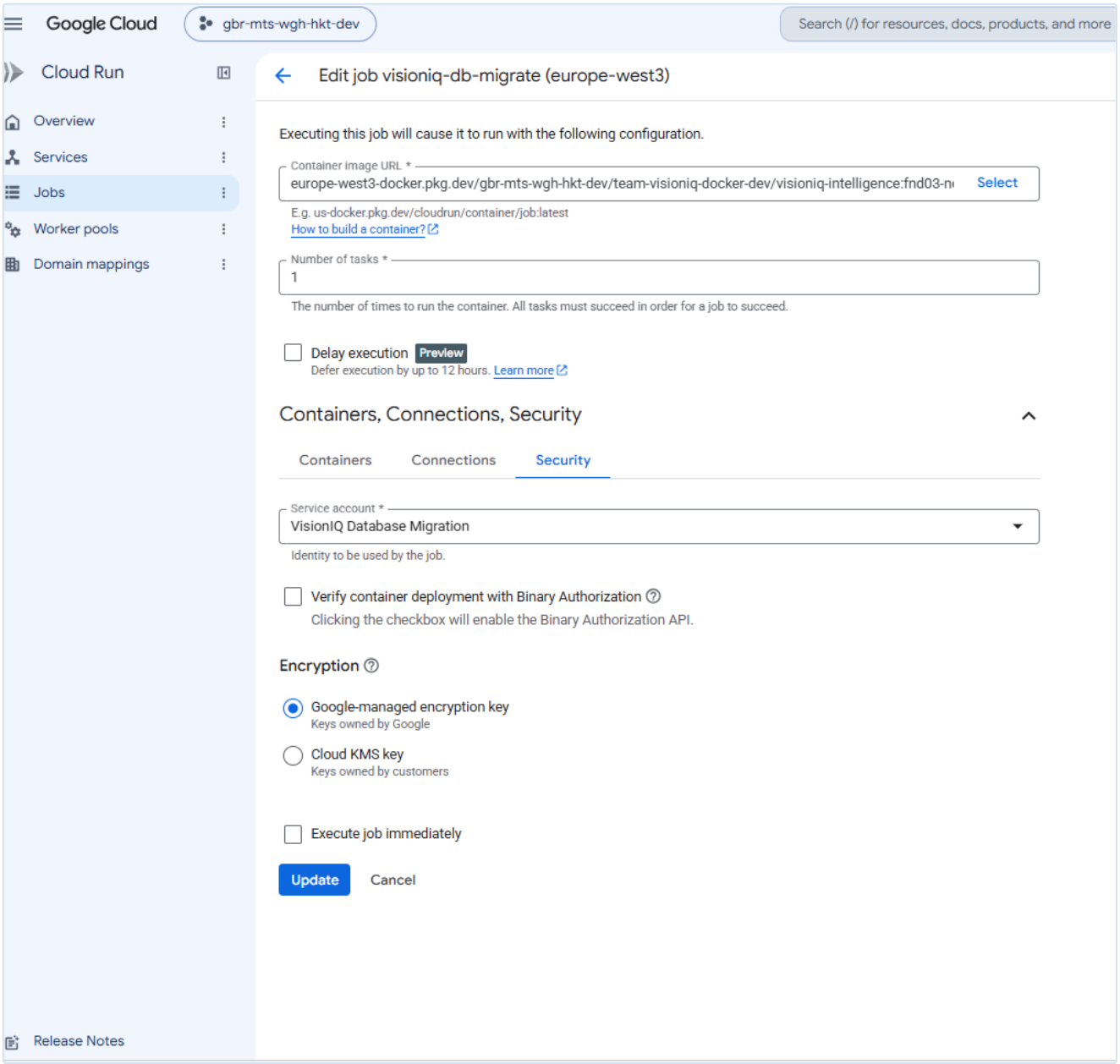


Observed precisely in the editor:

- **Connect to a VPC for outbound traffic:** selected.
- **Use Serverless VPC Access connectors:** selected, rather than **Send traffic directly to a VPC**.
- **Connector shown:** `vpc-con` under network `vpc-mts-wgh-hkt-mtswghhkt-dev`.
- **Traffic routing:** `Route only requests to private IPs to the VPC` selected (not route all traffic).
- **Cloud SQL connections:** no connection listed in that panel; this does not rule out connection to the private IP `10.102.16.8:5432` from application code.
- **Update button:** visible. This is an editable page and can contain changes that have **not been saved**.

SUCCESSFUL EXECUTION VERIFIED: The **Networking tab for** `visioniq-db-migrate-ngnk4` directly confirms connector `vpc-con` and private-only egress during the successful 08:27 execution; the later saved generation 10 YAML independently confirms the same connector. The **12:06 job update caveat no longer applies to identifying the successful execution's networking method**, although the precise sequence of earlier Direct VPC/connector experiments remains unverified. See CR-08 in Section 4.

CR-05 - Job workload identity and encryption



Observed: Friendly-name service account selection **VisionIQ Database Migration**, Google-managed encryption key selected; **Verify container deployment with Binary Authorization** unchecked. The editor displays a friendly service-account label, not the full email; the historical record identifies the intended principal as `visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`. Confirm the full identity through a read-only job description rather than treating the label alone as an exact principal proof.

What the screenshots do and do not establish

Established by the submitted screenshots	Not established by these editor screenshots
Job name and visible project/region	Whether the user clicked **Update** or saved anything
Container image tag and migration command displayed	Immutable image digest or deployment commit
Ten non-secret environment variables displayed	Whether those values were used for successful execution
Three Secret Manager **bindings** selecting `latest`	Secret values, secret IAM policy and actual runtime access logs

Established by the submitted screenshots	Not established by these editor screenshots
Saved job YAML and successful execution Networking **both confirm** `vpc-con`	Exact changes across failed executions and historical Direct VPC investigation
Friendly service account selection	Full deployed identity and exact IAM grants

Read-only confirmation (no job execution):

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION = 'europe-west3'
gcloud run jobs describe visioniq-db-migrate --project=$PROJECT --region=$REGION --format=yaml
# In the output inspect only the VPC connector/networkInterfaces, serviceAccount, image and command sections.
# Redact secret/environment values and internal identities before sharing anything publicly.

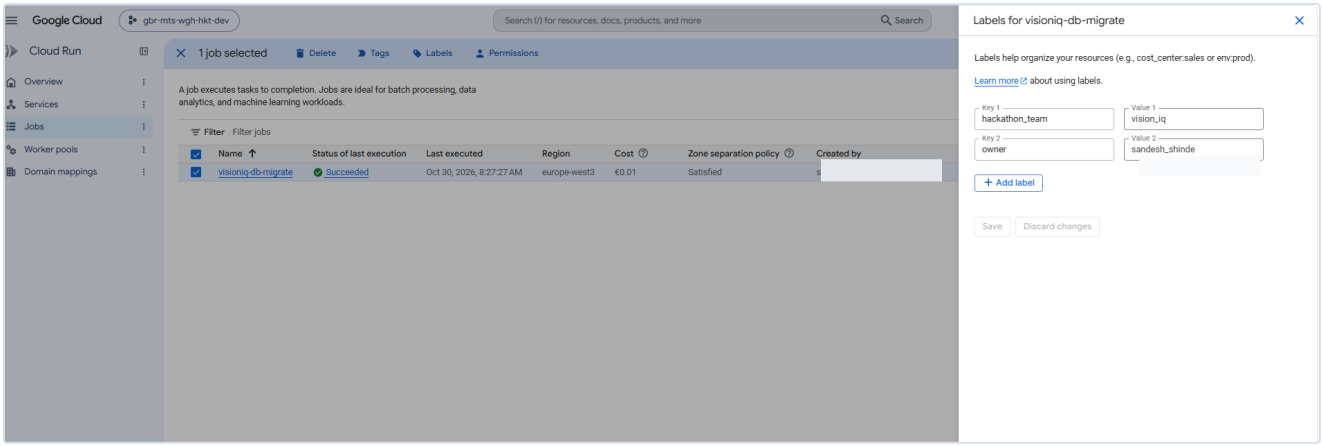
gcloud run jobs executions describe visioniq-db-migrate-ngnk4 --project=$PROJECT --region=$REGION --format=yaml
```

Networking evidence complete: The saved Job YAML, successful execution status **and execution-specific Networking screenshot** have been supplied. Next collect **sanitized logs** from a representative failed execution and from the successful execution to identify each failure's actual cause; the history list alone proves status, not errors.

Saved YAML, labels and execution history

Evidence sources: User supplied the full Cloud Run Job YAML and two Cloud Console screenshots: **(a)** Job list with Labels editor; **(b)** Job details/History with the successful execution selected. The two screenshot copies embedded below have personally identifying owner/creator/executor text obscured for sharing; the original evidence should remain restricted. The YAML has been summarized and excerpted without personally identifying annotations or secret values.

CR-06 - Labels and resource organization



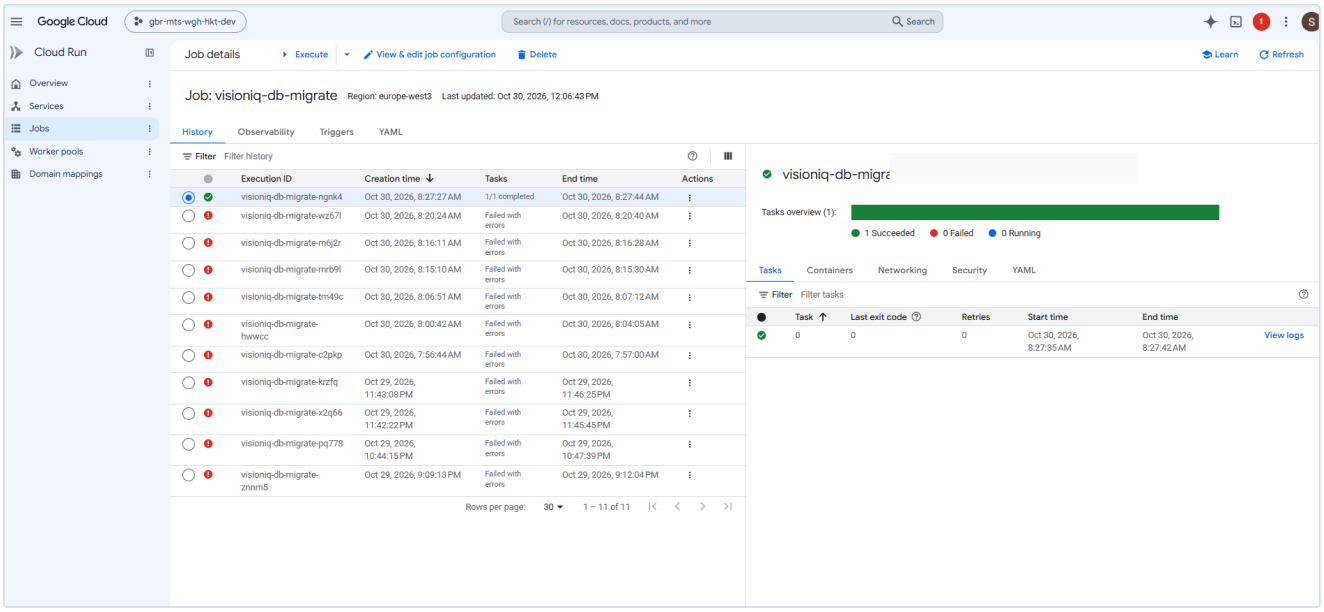
The user confirmed applying **resource labels** for distinguishing VisionIQ resources. The saved YAML shows the same application labels at the Job and template levels:

Label key	Recorded value in shared guide	Purpose
`hackathon_team`	`vision_iq`	Categorize VisionIQ hackathon resources.
`owner`	**Configured; value omitted in shared guide**	Ownership/accountability identification.

Label key	Recorded value in shared guide	Purpose
Platform-generated labels	Managed automatically; see original YAML	Region, platform flags and other system metadata.

Guideline for future resource creation: Apply the same team label to authorized VisionIQ resources where supported (Cloud Run, Artifact Registry, Cloud SQL, etc.), but **do not assume these labels exist on other resources until verified**. Owner details should remain in restricted corporate inventory, not public PDFs.

CR-07 - Full job execution history and successful task



The job **History** screen lists **11 executions**: **10 failed** and **one successful**. The selected execution `visioniq-db-migrate-ngnk4` has **1/1 completed**, **1 succeeded**, **0 failed**, **0 running**, **exit code 0**, **zero retries**. The screenshot provides outcome and time, *not the underlying exception for any failed execution*.

Date (Console, IST)	Execution ID	Start time (IST)	Recorded result
29 Oct	`visioniq-db-migrate-znnm5`	9:09:13 PM	Failed
29 Oct	`visioniq-db-migrate-pq778`	10:44:15 PM	Failed
29 Oct	`visioniq-db-migrate-x2q66`	11:42:22 PM	Failed
29 Oct	`visioniq-db-migrate-krzfq`	11:43:08 PM	Failed
30 Oct	`visioniq-db-migrate-c2pkp`	7:56:44 AM	Failed
30 Oct	`visioniq-db-migrate-hwwcc`	8:00:42 AM	Failed
30 Oct	`visioniq-db-migrate-tm49c`	8:06:51 AM	Failed
30 Oct	`visioniq-db-migrate-mrb9l`	8:15:10 AM	Failed
30 Oct	`visioniq-db-migrate-m6j2r`	8:16:11 AM	Failed
30 Oct	`visioniq-db-migrate-wz67l`	8:20:24 AM	Failed
30 Oct	`visioniq-db-migrate-ngnk4`	8:27:27 AM	**SUCCESS; completed 8:27:44 AM**

Do not assign an error category to a specific execution without its logs. Historical notes mention network connectivity, password-policy and IAM failures; the history screenshot alone does not map those causes to execution IDs.

Saved job metadata: evidence and timing

Saved YAML field	Observed value or interpretation
Resource	`kind: Job`, name `visioniq-db-migrate`
Region	`europe-west3`
`generation` / `observedGeneration`	`10` / `10`
`creationTimestamp` (UTC)	`2026-10-29T15:01:44.128569Z` = **29 Oct, 8:31:44 PM IST**
`lastUpdatedTime` (UTC)	`2026-10-30T06:36:43.353821Z` = **30 Oct, 12:06:43 PM IST**
`latestCreatedExecution`	`visioniq-db-migrate-ngnk4`
Execution creation (UTC)	`2026-10-30T02:57:27.828349Z` = **30 Oct, 8:27:27 AM IST**
Execution completion (UTC)	`2026-10-30T02:57:44.460915Z` = **30 Oct, 8:27:44 AM IST**
Execution status	`EXECUTION_SUCCEEDED`
`executionCount`	11
`Ready`	`True` (job is configured/ready; not a continuously running API)
Template	`run.googleapis.com/execution-environment: gen2`
Network	`run.googleapis.com/vpc-access-connector: .../connectors/vpc-con`
Egress	`run.googleapis.com/vpc-access-egress: private-ranges-only`
Execution tasks	`taskCount: 1`, `parallelism: 1`, `maxRetries: 0`
Limits	`cpu: 1000m`, `memory: 1Gi`, `timeoutSeconds: 600`
Identity	Migration service account `visioniq-db-migrate@...` (full approved identifier in restricted cloud inventory)
Encryption	Google-managed key shown in security screenshot
Image and entrypoint	`visioniq-intelligence:fnd03-netretry`; `python -m db.cloud_manage all --instance ...`

Timestamp finding - essential for accuracy: The currently saved Job YAML was updated at **12:06 PM IST**, approximately **3 hours 39 minutes after** the successful execution at **8:27 AM IST**. Thus, its **generation 10 networking settings prove the current saved configuration, not necessarily the network path in the earlier successful execution**. Record the successful execution's **Networking/YAML** before changing the historical account of how connectivity was fixed.

Sanitized configuration excerpt from provided YAML

The following excerpt reflects the saved **generation 10** job only. Sensitive and identifying metadata, full passwords and unrelated platform fields are omitted. It is **documentation**, not a complete `gcloud` import manifest.

```
kind: Job
metadata:
  name: visioniq-db-migrate
  generation: 10
  labels:
    hackathon_team: vision_iq
    owner: '[redacted in shared guide]'
spec:
  template:
    metadata:
      annotations:
        run.googleapis.com/execution-environment: gen2
        run.googleapis.com/vpc-access-egress: private-ranges-only
        run.googleapis.com/vpc-access-connector: projects/gbr-mts-wgh-hkt-dev/locations/europe-west3/connectors/vpc-
con
    spec:
      parallelism: 1
      taskCount: 1
      template:
        spec:
          serviceAccountName: visioniq-db-migrate@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com
          maxRetries: 0
          timeoutSeconds: 600
status:
  observedGeneration: 10
  executionCount: 11
  latestCreatedExecution:
    name: visioniq-db-migrate-ngnk4
    completionStatus: EXECUTION_SUCCEEDED
```

Read-only verification: execution networking - completed

COMPLETED 30 Oct: User supplied the **Networking tab on the successful execution `visioniq-db-migrate-ngnk4`**, captured separately from the latest job YAML. The screenshot confirms the networking mode *at execution*, not merely the later saved job generation. A supplementary read-only command is available for future audits:

```
$PROJECT='gbr-mts-wgh-hkt-dev'
$REGION='europe-west3'
gcloud run jobs executions describe visioniq-db-migrate-ngnk4 --project=$PROJECT --region=$REGION --format=yaml
```

The networking proof is complete. **Remaining evidence requests** are *sanitized task logs* from the successful execution and the most recent failure (`visioniq-db-migrate-wz67l`). They are required to assign exact root causes to failed runs, not to confirm the successful VPC route. No job should be rerun, updated or otherwise modified just for evidence collection.

Execution-level network proof

Refer to Section 4 for the successful execution Networking screenshot (CR-08). The Cloud Run job is a **one-shot administrative batch workload**, not a Cloud Run HTTP service. No .NET or Python application Cloud Run Service was deployed at the time of this evidence.

9. Cloud Run Services — Future Deployment

Current status: NOT DEPLOYED. The existence of the migration Cloud Run Job does not imply a .NET API or Python Intelligence Cloud Run Service is running in Google Cloud.

Application-service inventory verification — 30 October 2026

`gcloud run services list --project=$PROJECT --region=$REGION` returned **five services in the shared project**, but **no `visioniq-api` or `visioniq-intelligence` service**. These five entries are existing non-VisionIQ workloads and are intentionally not enumerated in this project-specific document. A Cloud Run **Job** named `visioniq-db-migrate` is not a Cloud Run **Service**.

This establishes that the intended VisionIQ application services were **not yet present in the inspected region**. It does not rule out similarly named services in other regions or projects. Deploy the application services only through an approved pipeline and with service-identity/network settings explicitly reviewed.

Pipeline source prepared — no cloud deployment yet (30 October 2026)

Code for two separate Cloud Build stages was committed to the public VisionIQ FND-04 feature branch and proposed for review as **draft PR #2**: FND-04 Cloud Build preparation. The authorized corporate source has **not** yet been connected, and no Google Cloud images or Cloud Run application services were deployed by this work.

Version-controlled file	Function	Expected privilege boundary
`cloudbuild/build-images.yaml`	Build .NET API and Python Intelligence images and push to existing `team-visioniq-docker-dev` Artifact Registry with approved 40-character Git commit tags	Dedicated build service account; Artifact Registry write, Cloud Logging write; no DB secret access
`cloudbuild/deploy-services.yaml`	Separate, manually approved Cloud Run deployment, using already-built tags and numeric Secret Manager versions	Dedicated deploy service account with reviewed Run deploy permissions and runtime-SA impersonation (`iam.serviceAccountUser`), not the DB migration SA
`cloudbuild/README.md`	Corporate GitHub connection, trigger setup, access boundaries, verification, exceptions and rollback runbook	No passwords, GitHub tokens or service-account keys

Proposed deployment defaults, not observed runtime settings: app services `visioniq-api` (port 5100) and `visioniq-intelligence` (port 8000); same approved `visioniq-runtime` Google identity; independent PostgreSQL logins; secret versions must be specified numerically; `vpc-con` with `private-ranges-only` egress; internal ingress; IAM authentication required; min instances 0 and max instances 2. A textual approval guard prevents accidental invocation but is **not a replacement for corporate IAM permissions and an approval-required trigger**.

Current blockers (updated later on 30 Oct): `visioniq-build-dev` connection creation was attempted twice and failed with Cloud Build service-agent Secret Manager permission errors; the last listed regional connection inventory was empty, and no trigger or app deployment is confirmed. The corporate GitHub connection requires an approved installation/admin path; build/deploy service accounts must be created and granted least privilege; corporate acceptance of the shared application identity, ingress, budget and network policies remains outstanding. The two applications' current `/health` endpoints do **not** test PostgreSQL connectivity or cross-service authentication. No actual deployment success is claimed.

When application services are deployed, extend this chapter in place. Include a subsection **per service** with:

Required field	What to record after deployment
Resource identity	Service name, project, region and environment

Required field	What to record after deployment
Artifact	Image URI and immutable digest, revision/commit and source Dockerfile
Runtime security	Service account, ingress authentication, IAM bindings, secrets
Network	VPC/connector or Direct VPC setting, egress, Cloud SQL access
Configuration	Non-secret environment variables, CPU, memory, concurrency, timeout, min/max instances
Validation	Service URL (if approved), authenticated smoke test, database query/readiness, logs
Operations	Deploy/redeploy command, rollback, cost and monitoring
Evidence	Sanitized Console captures and successful test output

Do not fill this table with invented production values. Separate application readiness from mere HTTP `/health` liveness.

Deployment deferred — user-story status and restart conditions (30 October 2026)

Explicit project decision: Leave **all FND-04 user stories OPEN**; do not close them based on prepared pipeline code, IAM grants or successful earlier migration work. Defer the Cloud Run application deployment until the organization can provide an approved repository integration or equivalent secure pipeline, appropriate deployer/build permissions and security sign-off. FND-03 remains separately complete based on its existing evidence.

Capability	State at pause	Evidence needed to resume / close
Public FND-04 source branch and draft PR #2	Prepared; four public CI jobs passed	Corporate code review, branch propagation and approved merge
Corporate GitHub ↔ Cloud Build connection `visioniq-build-dev`	**Creation failed twice**	Approved integration, successful connection state and repository link
Cloud Build triggers / automated `main` build	**Not created or executed**	Build trigger/run ID, approved identity, SHA-tagged pushed images
Application Cloud Run services	**Not deployed**	Two service revisions, exact digests, auth and health evidence
Private Cloud SQL access from each application	**Not tested**	Authenticated database queries from both services, restricted role checks
Workload identity separation / test-runner secret-denial	**Not accepted**	Security evidence or approved documented exceptions
Scheduler, AI/model and code-execution identities	**Not implemented/verified by this work**	Relevant acceptance evidence from subsequent stories

Operational handoff: Preserve this guide with FND-04 OPEN, the logs for failed operations and the IAM/Troubleshooter evidence. Before retrying, ask an authorized administrator to evaluate project-level Secret Manager Admin grants and Principal Access Boundary visibility. Review/revoke the unused GitHub PAT stored in `visioniq-build-token-dev` if that integration path is no longer needed. Resume deployment with sufficient time for Cloud SQL, service-to-service authentication and end-to-end demo tests.

10. Cloud Scheduler — Future Integration

Current status: No implemented VisionIQ Cloud Scheduler configuration has been verified. Permissions to use Cloud Scheduler were requested; an access request is not evidence that a job, schedule or trigger was created.

When Cloud Scheduler is introduced, record each job here: job ID, project/region, cron and timezone, HTTP/Pub/Sub target, service account and authorization, retries/attempt deadline, enabled state, target dependency, example payload without secrets, testing results, monitoring and disable/rollback steps. Add the new Console screenshots to the evidence section and update the change log. The complete scheduler documentation must remain in this master guide.

11. Other Cloud Products — Future Integration

The following are **future placeholders**, not a claim of provisioned resources:

Google Cloud product	Possible role	Current documented status	Future section requirements
Cloud Storage	Approved object storage if required for ingestion or generated artifacts	No VisionIQ bucket verified	Bucket ID, location, retention, IAM, encryption, data classification, lifecycle
Pub/Sub	Event-driven background processing when designed	No topic/subscription verified	Names, publishers/subscribers, retry/dead letter, auth, monitoring
Vertex AI	Gemini model inference when integrated into VisionIQ	No application inference verified by this infrastructure record	Project, models/locations, service account, quotas, data access, observability
Cloud Build	Application image build and approved release pipeline	**Code prepared in FND-04 public branch/draft PR #2; 0 regional connections and triggers as last inspected. No cloud execution yet.**	Corporate repository connection, reviewed triggers, distinct build/deploy identities, logs, release approvals
Other services	New approved services	Not yet documented	Add a service chapter with configuration, commands, evidence and changes

Cloud SQL Console integration toggles being enabled do not establish that a VisionIQ AI service was deployed or is functional. An optional curated knowledge export is not an FND deployment prerequisite.

12. Monitoring, Logging and Cost Management

Existing observed controls

Cloud SQL Query Insights was shown enabled; backups and PITR were also configured. These do not prove that application-level alerting, cost budgets or restoration drills are working. Cloud Run Jobs offer execution history, task exit status and logs; the recorded job shows one successful execution after ten failed executions.

Monitoring record to maintain

Area	What should be recorded
Cloud SQL	DB state, connections, CPU/memory/disk, backup results, PITR/ restore exercise, query latency
Cloud Run Jobs	Execution ID, date, task count, exit code, image digest, private network and job logs
Artifact Registry	Image retention, digests, stale image cleanup, storage costs
Secret Manager and IAM	Rotation, access review, policy changes and audit logs
Network	VPC connector health/capacity, routing and egress costs
Cost management	Billing project, labels, budgets/alerts, forecast/cost attribution

Recorded execution cost: the Console job-list screenshot displayed approximately €0.01 for the last job execution. This is a per-execution UI observation, not the monthly infrastructure cost of Cloud SQL, VPC connector or image storage. Costs must be checked using actual billing data before quoting a budget.

13. Operations, Verification and Safe Changes

Read-only infrastructure inventory

Where: PowerShell with an approved corporate Google Cloud identity and access to project metadata. These commands **inspect only**; do not treat successful output as proof of application-level connectivity.

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION  = 'europe-west3'
$DB      = 'visioniqdev'

# Check selected identity/project; avoid accidental operations against another project.
gcloud auth list
gcloud config get-value project

# Cloud SQL instance / database inventory.
gcloud sql instances describe $DB --project=$PROJECT --
format='yaml(name,state,region,databaseVersion,connectionName)'
gcloud sql databases list --instance=$DB --project=$PROJECT

# Cloud Run migration-job configuration.
gcloud run jobs describe visioniq-db-migrate --project=$PROJECT --region=$REGION

# Existing private network peering, allocation and egress-firewall metadata.
gcloud compute networks peerings list --network=vpc-mts-wgh-hkt-mtswghhkt-dev --project=$PROJECT
gcloud compute addresses list --global --filter='purpose=VPC_PEERING' --project=$PROJECT --
format='table(name,address,prefixLength,network)'
gcloud compute firewall-rules list --project=$PROJECT --filter='network:vpc-mts-wgh-hkt-mtswghhkt-dev AND
direction:EGRESS' --sort-by=priority --format='table(name,priority,denied,destinationRanges,disabled)'

# The known successful job execution (describe, not execute).
gcloud run jobs executions describe visioniq-db-migrate-ngnk4 --project=$PROJECT --region=$REGION
```

Expected: Cloud SQL details match Section 3; job configuration matches Section 8; the execution reports success. If `PERMISSION_DENIED`, seek required viewer authorization rather than trying to bypass corporate controls.

Repeatable pre-deployment cloud inventory — PowerShell (verified commands)

Where: Windows PowerShell with Google Cloud CLI authenticated to the approved corporate account. All commands below are **read-only** and were successfully executed on **30 October 2026**. They check the cloud environment and do not deploy any workloads. Other unrelated resources may appear in unfiltered listings; this runbook does not authorize modification of them.

```
$PROJECT = "gbr-mts-wgh-hkt-dev"
$REGION = "europe-west3"

# Identity and selected project (do not paste personal email outputs in documentation)
gcloud auth list
gcloud config get-value project

# Cloud Run APPLICATION services (not Cloud Run Jobs)
gcloud run services list `
  --project=$PROJECT `
  --region=$REGION

# Artifact Registry repositories for this region
gcloud artifacts repositories list `
  --project=$PROJECT `
  --location=$REGION

# Existing VisionIQ service accounts
gcloud iam service-accounts list `
  --project=$PROJECT `
  --filter="email:visioniq"

# Cloud Build triggers in this region only
gcloud builds triggers list `
  --project=$PROJECT `
  --region=$REGION

# Shared private VPC connector used by the migration job
gcloud compute networks vpc-access connectors describe vpc-con `
  --project=$PROJECT `
  --region=$REGION
```

Expected from the 30 October observation: correct project; no VisionIQ application services in `europe-west3`; VisionIQ Docker repository present; two enabled VisionIQ service accounts; zero visible regional Cloud Build triggers; `vpc-con` READY. Record actual future outputs rather than assuming they remain unchanged.

CLI flag correction: use `--region` for `gcloud run services list`, not `--LOCATION`. The latter was rejected by the installed CLI.

Follow-up before deploying app services: validate effective runtime identity IAM, Artifact Registry read/deploy permissions, Secret Manager bindings and least privilege, network authorization to Cloud SQL, budget configuration, and pipeline availability (including any corporate runner restrictions). Existing shared resources must not be modified as part of a VisionIQ-only verification.

IAM, secrets and Cloud Build pre-deployment inspection — PowerShell (verified and proposed)

Verified 30 October 2026 (read-only): the commands in the first block below were executed and produced the results summarized in the IAM chapter. These commands do not change access or deploy workloads.

```
$PROJECT = "gbr-mts-wgh-hkt-dev"
$REGION = "europe-west3"
$RUNTIME_SA = "visioniq-runtime@$PROJECT.iam.gserviceaccount.com"
```

```
# Confirm required APIs are enabled (the five listed APIs were returned)
gcloud services list --enabled `
  --project=$PROJECT `
  --format="value(config.name)" |
  Select-String "cloudbuild.googleapis.com|run.googleapis.com|artifactregistry.googleapis.com|
secretmanager.googleapis.com|iam.googleapis.com"

# Cloud Build 2nd-gen host connections (observed: 0 in europe-west3)
gcloud builds connections list --project=$PROJECT --region=$REGION

# Direct project-level IAM bindings for the runtime principal (observed: no rows)
gcloud projects get-iam-policy $PROJECT `
  --flatten="bindings[].members" `
  --filter="bindings.members:$RUNTIME_SA" `
  --format="table(bindings.role)"

# Direct IAM bindings ON the service-account resource (observed: etag only)
gcloud iam service-accounts get-iam-policy $RUNTIME_SA --project=$PROJECT
```

Executed and verified on 30 October 2026 (read-only): review only secret **names and IAM metadata**. Never access or display secret payloads, version contents, access tokens or key files. The following per-secret commands may display principal identities; redact personal information before sharing or attaching evidence.

```
$PROJECT = "gbr-mts-wgh-hkt-dev"

# Secret names only, without retrieving any secret values
gcloud secrets list --project=$PROJECT --format="value(name)"

# Inspect direct IAM bindings on ONLY the two application DB secrets
foreach ($secret in @("visioniq-db-api-password", "visioniq-db-intelligence-password")) {
    Write-Host "Inspecting IAM metadata for $secret"
    gcloud secrets get-iam-policy $secret `
      --project=$PROJECT `
      --format="yaml(bindings)"
}
```

Historical snapshot (before IAM update): Both policies originally listed only the migration service account. The shared runtime identity was **subsequently added to both** and each update returned the new bindings. For the *current* configuration, repeat the read-only `get-iam-policy` commands above; no secret values are required. Deployment identity permissions and the corporate Cloud Build connection are still unresolved.

Executed change — runtime Secret Accessor grants (30 October 2026)

State-changing commands already executed successfully; retained only for reproducibility and audit, not to be run again automatically:

```
$PROJECT = "gbr-mts-wgh-hkt-dev"
$RUNTIME = "serviceAccount:visioniq-runtime@$PROJECT.iam.gserviceaccount.com"

# Existing app-password secret (migration grant remains intact)
gcloud secrets add-iam-policy-binding visioniq-db-api-password `
  --project=$PROJECT `
  --member=$RUNTIME `
  --role="roles/secretmanager.secretAccessor"

# Intelligence password secret (migration grant remains intact)
gcloud secrets add-iam-policy-binding visioniq-db-intelligence-password `
  --project=$PROJECT `
  --member=$RUNTIME `
  --role="roles/secretmanager.secretAccessor"
```

Verification (read-only, safe to repeat):

```
$PROJECT = "gbr-mts-wgh-hkt-dev"
foreach ($secret in @("visioniq-db-api-password", "visioniq-db-intelligence-password")) {
    gcloud secrets get-iam-policy $secret --project=$PROJECT `
      --format="yaml(bindings)"
}
```

Rollback if explicitly authorized: The inverse `gcloud secrets remove-iam-policy-binding` operation with the same `--member` and `--role` removes the **shared runtime** binding for each approved secret. Do **not** remove access from deployed applications until a replacement identity/secret-access path is verified. This IAM rollback is independent of the successful Cloud Run database migration job.

Controlled migration execution — changes database state

Important: This is a change operation against an actual development database. Execute **only** with authorized deployment permissions and a reviewed migration/image. Do not use the local `docker compose` migration tool for Cloud SQL.

Where: PowerShell in an approved deployment environment. Confirm cloud project, region, image, job configuration, identity, private VPC, current Alembic head, and a recovery plan before execution.

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION  = 'europe-west3'

# Inspect before making changes.
gcloud config get-value project
gcloud run jobs describe visioniq-db-migrate --project=$PROJECT --region=$REGION

# ONLY after approval, and only when a new reviewed migration needs applying:
gcloud run jobs execute visioniq-db-migrate --project=$PROJECT --region=$REGION --wait
```

Success: Job reports success; logs show private TCP connectivity, database-role/bootstrap success, and pgvector/full-text/ACL verification. **If it fails:** do not assume partial migration/role writes rolled back; inspect logs, repair the cause, verify actual database state, and rerun only after review. Do not reset the Cloud SQL instance or destroy the database to recover.

SQL Studio read-only evidence (verified 30 Oct):

```
SELECT version_num FROM public.alembic_version;
SELECT schema_name FROM information_schema.schemata
 WHERE schema_name IN ('workflow','evidence','knowledge') ORDER BY schema_name;
SELECT extname FROM pg_extension WHERE extname = 'vector';
SELECT to_tsvector('english','historical defect regression')
       @@ plainto_tsquery('english','defect') AS fulltext_ok;
```

Expected: `20261029_0001`; three schemas; `vector`; `fulltext_ok=true`. The migration `verify` is stronger than merely inspecting ACLs because it attempts permitted and forbidden writes through both runtime DB identities. Do not paste the actual password into manual SQL; if administering role `visionIQ` through manual SQL, quote its case-sensitive name exactly: `"visionIQ"`.

Source-controlled migration code and responsibilities

Authoritative development target: the approved corporate private **Defect-Intelligence** repository (code push confirmed; merge pending/unverified). A temporary personal development bridge was used during implementation; do not treat that bridge as the team source of truth or publish its account identifiers in this guide.

Repository path	Responsibility
`db/manage.py`	Safe local-only migration (`migrate`), bootstrap (`roles`), verification (`verify`) and combined (`all`). Refuses cloud target.
`db/cloud_manage.py`	Guarded Cloud SQL runner; requires explicit `DB_TARGET=cloud` , matching project/instance, private IPv4/5432, unique credentials and `VISIONIQ_CLOUD_RUN_MIGRATIONS=approved` .
`db/migrations/env.py`	Alembic migration context. Cloud runner sets `VISIONIQ_MIGRATION_CONTEXT=cloud` only around migration. `target_metadata=None` until ORM models are registered.
`db/alembic.ini` + Alembic versions	Single ordered migration history. Confirmed database revision `20261029_0001` .
`db/Dockerfile`	Dedicated one-shot migration image. Copies `db/` and installs the one root `requirements.txt` .
`src/VisionIQ.Intelligence/Dockerfile`	Python runtime image, deliberately **does not copy** `db/` migration code.
`src/VisionIQ.Api`	.NET API with dedicated database credential configuration.
`docker-compose.yml`	Local PostgreSQL + migration one-shot + .NET/Python services. API containers wait for `db-migrate` to exit successfully.
`.env.example`	Safe-to-commit configuration names only. `.env` holds private local values and must not be pushed.
`requirements.txt`	Single Python dependency file for runtime, migrations and tests (hackathon simplification).
`.github/workflows/pr-validation.yml`	Passed on the public bridge; intentionally removed/deferred on corporate GitHub because hosted runners are unavailable.

Current migration actions:

1. `migrate` : Alembic upgrade to head; revision `20261029_0001` produced three schemas and `vector` extension.
2. `roles` : create/update limited API and Intelligence DB roles; revoke `PUBLIC` permissions; grant intended schema/table/sequence access; configure default privileges for future migration-created objects.
3. `verify` : check `vector` extension, PostgreSQL full-text match, authorized writes, cross-schema write denial, reads and rejected DDL; clean up test tables during normal completion/errors.
4. `all` : run steps 1–3. Both local Compose and explicit Cloud Run job use the same migration/verification foundation, with different credential selection and safety guards.

Known technical limitations: `bootstrap` uses autocommit and is rerunnable rather than fully atomic; a forced kill during verification can leave temporary `fnd03_probe_*` tables; Alembic autogenerate is unavailable until mapped model metadata is wired; TCP reachability probe is executed before every Cloud runner action. See Section 14. These do **not** invalidate the successful verification evidence.

Safety principles

- Do not run a Cloud Run job merely to collect documentation screenshots or logs.
- Do not delete Cloud SQL instances, local Docker volumes, secrets or service accounts to recover from an investigation.

- Confirm project, region, IAM authorization, expected migration head, image digest and backup/recovery plan before a write operation.
- Maintain a clear distinction between **reference example commands** and actual commands recorded from history.
- Keep local everyday setup in its own companion guide; this chapter concerns Google Cloud operations.

Cloud Build pipeline activation — unsuccessful connection attempts; deployments NOT executed

Status: PREPARED / DEFERRED. The files `cloudbuild/build-images.yaml`, `cloudbuild/deploy-services.yaml`, and `cloudbuild/README.md` exist in the public FND-04 branch (draft PR #2). Two actual Cloud Build connection creation attempts were executed in Google Cloud and **both failed**. No successful VisionIQ source connection, trigger, application image build through this new pipeline, or Cloud Run application service deployment is established by the evidence.

The private corporate repository is on `github.com`. Corporate GitHub App installation/authorization access was unavailable. The operator created a manual GitHub authorization-token secret in an attempted alternative; this **did not produce a connection**. Do not use unrelated corporate GitLab integration secrets or unsecured/personal-token bypasses. An administrator-approved build/deployment path is required.

Read-only regional checks (verified previously; rerun as needed):

```
$PROJECT = "gbr-mts-wgh-hkt-dev"
$REGION  = "europe-west3"
gcloud builds connections list --project=$PROJECT --region=$REGION
gcloud builds triggers list --project=$PROJECT --region=$REGION
gcloud iam service-accounts list --project=$PROJECT --filter="email:visioniq"
# Secret VERSION METADATA only; never access or print actual password values.
gcloud secrets versions list visioniq-db-api-password --project=$PROJECT
gcloud secrets versions list visioniq-db-intelligence-password --project=$PROJECT
```

Actually executed (twice) and failed — historical record:

```
$PROJECT = "gbr-mts-wgh-hkt-dev"
$REGION = "europe-west3"
gcloud builds connections create github visioniq-build-dev `
  --project=$PROJECT `
  --region=$REGION
```

The two Google Cloud operations failed with **code 9** and a message that the Cloud Build P4SA could not assert Secret Manager permissions. The specific denied permission reported was `secretmanager.secrets.create`; `secretmanager.secrets.setIamPolicy` was also named as a required permission. Operation identifiers (as provided in operator logs):

```
projects/gbr-mts-wgh-hkt-dev/locations/europe-west3/operations/
operation-1793355871367-65f0c33ab316b-08425502-133a7966
projects/gbr-mts-wgh-hkt-dev/locations/europe-west3/operations/
operation-1793356362062-65f0c50ea9adf-92f2b395-4bf67ba0
```

After the first failure, read-only IAM inspection showed five project roles on the service agent (Section 7). A connection list still returned `Listed 0 items`. Retrying the identical command failed again. Policy Analyzer displayed the `secretmanager.secrets.create` allow grant, but Policy Troubleshooter reported an **UNKNOWN effective outcome**. **No successful GitHub connection or deployed application follows from these attempts.**

Manual-token alternative (partially performed): The operator reported creating Secret Manager secret `visioniq-build-token-dev` and granting the Cloud Build P4SA Secret Accessor on that secret through the UI. The GitHub Cloud Build App could **not** be installed/authorized with the available corporate rights, so **no manual-token connection command was run** and no GitHub App installation ID was obtained. The token is not needed for direct local-source Cloud Build submissions and should be reviewed for revocation/retirement while FND-04 is deferred.

Alternative considered but not executed: direct local-source submission through `gcloud builds submit . --config=cloudbuild/build-images.yaml` could avoid the GitHub App integration, but would still require authorized build identities, correct source revision, image publishing rights and an approved deployment release. Cloud Run's 'Connect repository' UI may provision GitHub/Developer Connect integration behind the scenes; it is not an automatic workaround for organization GitHub authorization policies. No such source build or Cloud Run app deployment was reported.

Trigger model: (1) Build on reviewed corporate `main` pushes using a dedicated build identity, a SHA-tagged image for each service, without runtime secrets; (2) a **separate manual, approval-required deploy trigger**, using a dedicated deploy identity and numeric API/Intelligence secret versions. Runtime service identity remains `visioniq-runtime` pending a documented acceptance of shared access. Use Cloud Build's `CLOUD_LOGGING_ONLY` log destination for user-managed build identities after granting appropriate Cloud Logging permissions.

Deployment safety: Deployment is configured to require an explicit release approval, both images, valid secret-version identifiers, and reviewed ingress/network IAM policies. It does not execute Alembic or modify the successful `visioniq-db-migrate` job. Deploying the two services is not an atomic operation; record the previous image digests/revisions before changes. Verification must include authenticated `/health`, a real DB read per service, Cloud Run Invoker authorization, and a test-runner secret-access denial.

Audit after actual provisioning: Update Chapters 7, 9, 11, 12 and 17 with service-account names, IAM bindings, connection/trigger IDs, build and deployment execution IDs, image digests, Cloud Run revisions, exact timestamps, failure logs and sanitized screenshots. Until then keep status **PREPARED — NOT DEPLOYED**.

14. Troubleshooting and Known Technical Issues

Diagnosed issues, evidence and resolutions

Incident	User-visible symptom / evidence	Root cause or most likely boundary	Resolution / verification	Documented status
Cloud Build source connection `visioniq-build-dev`	Two `gcloud builds connections create github` operations returned code 9, `secretmanager.secrets.create` denied for Google-managed Cloud Build P4SA	Despite five project-level role bindings, Policy Analyzer showed grant while Troubleshooter result was UNKNOWN (conditional Allow; inaccessible PAB; Deny permits). GitHub App installation not approved/available	Stop retries; authorized org IAM review + GitHub integration approval; no deployed connection	**BLOCKED / FND-04 OPEN**
GitHub authorization-token secret for attempted workaround	`visioniq-build-token-dev` created through Console; secret-level Cloud Build Secret Accessor grant operator-confirmed	No App installation ID / no successful manual connection	Keep value confidential; revoke PAT and retire secret if unneeded under approved control	**UNUSED / cleanup review**

Incident	User-visible symptom / evidence	Root cause or most likely boundary	Resolution / verification	Documented status
GCP project switch/ access	`gcloud config set project ...` did not provide access	Corporate IAM project permissions absent	Use authorized account/project; request access; explicit project flags in commands	Corporate access still role-dependent
Service Networking inspection	`servicenetworking.services.get` denied	Viewing Service Networking peering API was not allowed	Compute peerings and allocated-range inspection performed; no duplicate peering created	Diagnostic workaround confirmed
Cloud migration could not reach private DB	Early Cloud Run executions failed; precise per-execution logs still not attached	VPC reachability and credentials were investigated	**Successful execution verified connector** `vpc-con`, private-IP-only egress; TCP probe passed	Connectivity RESOLVED; failure-to-fix chronology OPEN
Earlier Direct VPC notes versus successful connector	Earlier work mentions Direct VPC; **successful 08:27 execution Networking confirms** `vpc-con`	Direct VPC may have been tried or considered during earlier failed attempts	**Final successful route VERIFIED** `vpc-con`; historical Direct VPC attempt timeline needs original logs/ Audit records	Successful route RESOLVED; chronology OPEN
Cloud SQL login / password policy	Database authorization/ authentication errors	Admin username casing, credential mismatch and distinct password requirements	Preserve case-sensitive `"visionIQ"`; configure distinct admin/API/ Intelligence password secrets	RESOLVED for successful run
App and deployer IAM role grants	Some corporate IAM operations were denied; two app secret grants succeeded later	Shared project requires administrator-controlled access	DB application-secret access now verified at secret level; cloud build/deployer identities and effective privileges remain unverified	**PARTIAL / FND-04 OPEN**
Cloud runner image drift	New `db/Dockerfile` exists in source but Cloud Run had older image tag	Source hardening happened after successful deployed image	Record digest; deploy dedicated image only when approved for a new migration	PENDING next release
GitHub CI	Hosted runners disabled for private repository	Enterprise policy, not test failure	Private CI deferred; public bridge passed at recorded commit	DEFERRED
Local Docker password validation	`Use three different database passwords`	Reused credentials across distinct DB roles	Updated `.env` to distinct secrets; existing volume password retained	RESOLVED
Local Compose configuration missing (30 Oct)	`no configuration file provided: not found`	Command launched from a directory lacking Compose file or wrong branch	File discovery/branch validation documented in Docs 02-03; awaiting local command output	OPEN

Important: The exact early Cloud Run execution IDs, full error messages, timestamps and user-entered commands have **not** been recovered from the records. The incident journal describes observed categories;

it must not be mistaken for a byte-for-byte terminal transcript. Google Cloud Logging and sanitized PowerShell history can fill those gaps.

Read-only Cloud Run troubleshooting collection

```
$PROJECT='gbr-mts-wgh-hkt-dev'; $REGION='europe-west3'
gcloud run jobs executions list --job=visioniq-db-migrate --project=$PROJECT --region=$REGION
gcloud run jobs executions describe visioniq-db-migrate-ngnk4 --project=$PROJECT --region=$REGION
# Console: Cloud Run > Jobs > visioniq-db-migrate > Executions > [specific failed run] > Logs
```

For each failure, preserve: **symptom -> actual first error line -> resource/config state -> exact fix -> rerun result**. Redact secrets, private user identifiers, employee information and any broad environment dumps.

> **G-07 Failure screenshots:** Cloud Run failed execution list, one redacted failed log per root-cause category, and successful execution `visioniq-db-migrate-ngnk4`.

> **G-08 SQL Studio:** screenshots of Alembic version `20261029_0001`, three schemas and `vector` extension; SQL-query text and success row should be legible.

Technical risks and known limitations

Finding	Severity / action	Why it is not blocking the completed foundation
Alembic `target_metadata=None`	**Deferred until ORM models**	Handwritten migrations work; autogenerate needs mapped SQLAlchemy metadata later.
Bootstrap uses autocommit	**Hardening**	Failure partway through leaves partial grants; process is rerunnable. Review transaction boundaries and Cloud SQL role-DDL behavior before changing.
Abruptly killed `verify` may leave `fnd03_probe_*` tables	**Hardening**	Normal error/completion executes `finally` cleanup; add stale-probe cleanup before final production use.
TCP probe runs for `migrate`, `roles`, `verify` and `all`	**Minor optimization**	Some startup overhead, but useful for VPC startup diagnosis.
Migration image in Cloud Run older than new dedicated image source	**Before next migration release**	Existing execution passed; validate/ publish dedicated image for any new cloud migration.
Shared runtime Google SA for .NET and Python	**Proposed hackathon shortcut; FND-04 acceptance OPEN**	DB logins remain different, but cloud-secret IAM is shared. Obtain explicit corporate approval or separate runtime identities before closure.
SSL `require` rather than `verify-full`	**Security hardening**	TLS encrypted but hostname verification not established. Review organizational TLS policy.
IAM database auth flag enabled but SQL passwords used	**Documented design**	No claim that the runner uses IAM DB authentication.

Finding	Severity / action	Why it is not blocking the completed foundation
Existing corporate VPC reuse approval	**Open external dependency**	Network team confirmation requested; avoid unilateral changes.

Historical execution accuracy: The list of ten failed runs proves they failed, but not the exact cause of each run. Preserve sanitized failure logs to connect every individual failure to its diagnosis and correction. The successful execution used `vpc-con`; earlier Direct VPC settings belong to troubleshooting history only.

15. Infrastructure Implementation History

This historical register records infrastructure changes and implementation decisions without making any development milestone the organizing principle for the overall cloud document.

Historical creation, migration and release sequence

Date	Action / observation	Outcome
29 Oct 2026	Reviewed four Cloud SQL Configurations screenshots and recorded instance/ backup/network/security settings	**CONFIGURED** (visual evidence only at that time)
29–30 Oct 2026	Verified existing PSA / allocated private CIDR `10.102.16.0/20` and active network peerings; checked egress firewall output	**OBSERVED** via gcloud. Service Networking peering API read was permission-denied.
29–30 Oct 2026	Updated `visioniq-db-migrate` job for approved VPC network and subnet with `private-ranges-only` egress	**CONFIGURED** via CLI. Early job attempts failed; diagnosis and credential/network changes followed.
30 Oct 2026	Resolved PostgreSQL username/ password-policy and authentication issues using separate admin/API/ Intelligence credentials	**RESOLVED**, as confirmed by successful subsequent migration; no secrets recorded.
30 Oct 2026	Cloud Run execution `visioniq-db-migrate-ngnk4`	**SUCCEEDED**. TCP private endpoint reachable; role bootstrap and pgvector/ full-text/ACL verification passed.
30 Oct 2026	Cloud SQL Studio inspected `public.alembic_version`, `workflow`, `evidence`, `knowledge`, and extension `vector`	**VERIFIED**; recorded revision `20261029_0001`.
30 Oct 2026	Source hardening: dedicated runtime DB identity variables; separate migration/ runtime Dockerfiles; single `requirements.txt`	**IMPLEMENTED** on development branch.
30 Oct 2026	Public bridge GitHub Actions jobs (Python tests/lint, .NET 10, Docker image builds)	**PASSED** on commit `6f369af4c582eca0d2aa2928f7532fe8915af625`.
30 Oct 2026	Local Compose failed validation because database passwords were reused	**FIXED** by configuring three distinct credentials; later user reported all services healthy.

Date	Action / observation	Outcome
30 Oct 2026	Corporate private repository CI jobs all blocked before tests by hosted-runner enterprise policy	**DEFERRED**; workflow removed on user's private branch per decision. Not a code test failure.
30 Oct 2026	Two Google service accounts exist; IAM policy assignment attempt failed; access email sent to cloud team	**CONFIGURED / IAM PENDING**. No role assignment completion confirmed.
30 Oct 2026	Fresh-volume isolated local migration test supplied	**PENDING** result; see the companion Complete VisionIQ Setup guide.

Historical implementation boundary

The initial VisionIQ database foundation produced one Cloud SQL instance, one Artifact Registry repository, three secrets, two service accounts and one Cloud Run migration job. Existing corporate networking was reused. The .NET/Python Cloud Run application Services had not been deployed. Other lifecycle work belongs in subsequent changes to their respective service chapters.

Historical local/repository notes — retained for traceability only

Do not confuse local Docker, the corporate private-repository PR, public development-bridge CI or the local fresh-volume migration test with Google Cloud resource deployment. Their operational instructions belong in **Complete VisionIQ Setup** and **Run VisionIQ Locally**; these details are retained only insofar as they explain cloud migration packaging and historical troubleshooting.

16. Evidence and Screenshot Register

Screenshot and configuration evidence register

Each screenshot should be inserted next to the technical step it proves, not collected as unlabelled images at the end. Use a figure caption with resource, date, result and permitted redactions. An image from a later date should not be captioned as if it shows the exact original deployment state.

ID	What to capture	Screen/command	Why this is needed	Status
G-01	Cloud SQL instance/ Config overview (4 screens from 29 Oct if available)	Cloud SQL > visioniqdev > Overview/ Configurations	Original resource options, not guesswork	AWAITING IMAGE
G-02	VPC network, subnet, PSA/peering, allocated range	VPC network/Peering + IP addresses	Why private connectivity works	AWAITING IMAGE
G-03	Job Edit -> Connections and saved Job YAML	Cloud Run -> Jobs -> visioniq-db-migrate	Current saved connector `vpc-con` confirmed	**CAPTURED**
G-04	Registry repository + built image/digest	Artifact Registry	Distinguish built/ deployed images	AWAITING IMAGE
G-05	Job container command, image, tasks, limits	Edit job -> Containers	Container command and displayed settings, not execution result	**CAPTURED 30 Oct**

ID	What to capture	Screen/command	Why this is needed	Status
G-06	Job editor secret references and Security tab	Edit job -> Variables & Secrets / Security	Three secret **references** and friendly-name migration identity; no secret contents	**CAPTURED 30 Oct; policy verification PENDING**
G-07	Execution history: 10 failed, 1 succeeded	Cloud Run -> Jobs -> History	**Captured 30 Oct**, exact failure logs still needed	**CAPTURED HISTORY / LOGS PENDING**
G-08	Alembic, schemas and vector in SQL Studio	Cloud SQL > SQL Studio	Actual database proof	AWAITING IMAGE
G-09	IAM denied operation and cloud admin request status	Error screenshot, no personal email text	Explain outstanding blocker	OPTIONAL
G-10	Job saved YAML generation 10 + label keys	Cloud Run -> Jobs -> YAML	Confirms connector, identity, image, limits and job timestamps	**CAPTURED 30 Oct**
G-11	Successful execution Networking screenshot (CR-08)	Select execution `visioniq-db-migrate-ngnk4` -> Networking	**Proves successful job used `vpc-con`, connector subnet and private-IP-only routing**	**VERIFIED 30 OCT**

Screenshot capture policy: Crop to relevant panels; preserve full originals in a restricted evidence package. The five images in Section 8 are supplied Cloud Run job **edit-page** screenshots. They contain no plaintext password values. For future captures, hide bearer tokens, credentials, passwords, unrelated corporate content and individual identity details. Record capture date and what an image actually proves, not what one hopes it proves. Missing images remain PENDING rather than fabricated.

Already embedded in this handbook

- CR-01 container settings; CR-02 non-secret environment variables; CR-03 Secret Manager bindings; CR-04 Job editor networking; CR-05 security/identity.
- CR-06 labels; CR-07 historical execution list; CR-08 **successful execution-specific networking**.
- All image captures are sanitized for sharing. Missing Cloud SQL/Artifact Registry/IAM/Secret Manager inventory screens remain labelled as requested evidence, not supplied evidence.

Every future Google Cloud service section should cite the relevant screenshot here and preserve the configuration alongside explanatory text—not screenshots without interpretation.

17. Infrastructure Change Log

Update this single cloud infrastructure document whenever a Google Cloud resource is provisioned, modified, tested, retired or replaced. Document the effect in its service chapter and add an entry below. Revision numbers are metadata for history; they do not belong in the executive introduction or scope.

Date	Revision	Recorded change / evidence
29 Oct 2026	1.0	Initial Cloud SQL configuration from supplied settings.

Date	Revision	Recorded change / evidence
29 Oct 2026	1.1	Four Google Cloud Console screenshots reviewed; security, backups, maintenance, network and manual provisioning captured.
30 Oct 2026	**1.2**	Updated actual FND-03 Cloud Run migration/job configuration, PostgreSQL roles and schema policy, migration execution and SQL Studio verification, password-policy troubleshooting, latest code/image packaging, local Compose status, enterprise CI limitation, two service accounts, pending IAM/VPC access approvals, fresh-volume acceptance steps and FND-04 handoff.
30 Oct 2026	**1.3**	Consolidated historical command catalog, Direct VPC/connector explanation, incident journal, screenshot register and cross-references.
30 Oct 2026	**1.4**	Added five Cloud Run job editor screenshots (container, env, Secret Manager references, Connections, Security). Marked the `vpc-con` versus historically recorded Direct VPC discrepancy as OPEN; clarified editor state versus deployed/executed configuration.
30 Oct 2026	**1.5**	Added saved job YAML (generation 10), two redacted screenshots (labels and full history), 11 execution outcomes, execution timestamps and current `vpc-con` route.
30 Oct 2026	**1.6**	Added sanitized execution-specific Networking screenshot CR-08. **Resolved successful migration networking as Serverless VPC Access `vpc-con` / private-IP-only** with exact VPC and connector subnet. Retained Direct VPC as unverified historical troubleshooting, not final architecture.
30 Oct 2026	**1.7**	**Clarified FND-03 cloud scope:** created Cloud SQL, Artifact Registry, three Secret Manager secrets, two service accounts and one migration Cloud Run Job. Distinguished existing/ shared VPC and PSA from VisionIQ-owned resources, and explicitly marked .NET/Python Cloud Run application Services NOT deployed. Added one-stop read-only inventory commands and outstanding screenshot evidence.

Date	Revision	Recorded change / evidence
30 Oct 2026	**1.8**	**Verified read-only cloud environment inventory:** corporate CLI project selection, absence of VisionIQ Cloud Run application services in `europe-west3`, VisionIQ Artifact Registry repository and reported size, two enabled service accounts, no regional Cloud Build triggers, and READY `vpc-con` configuration/capacity. Added copy-paste PowerShell runbook, documented corrected `--region` flag, and kept shared resources out of VisionIQ ownership assertions. No cloud resources changed.
30 Oct 2026	**1.9**	**Read-only Cloud Build and IAM readiness evidence:** confirmed five enabled platform APIs; zero 2nd-gen Cloud Build connections visible in `europe-west3`; no direct project-policy row for `visioniq-runtime`; direct service-account policy returned only `etag: ACAB`. Clarified direct vs inherited and secret-level IAM, approved deployment identity vs runtime identity, and the next secret-metadata inspection commands. No cloud resources or permissions were changed.
30 Oct 2026	**1.10**	**Read-only Secret Manager verification (pre-change snapshot):** confirmed the three VisionIQ DB secret names, inspected API and Intelligence secret policies before the runtime grants, observed only the migration account as a direct accessor and recorded the shared-vs-separate identity decision. Excluded unrelated corporate secrets and all secret values. This inspection made no changes.
30 Oct 2026	**1.11**	**Two actual IAM updates:** Added the shared `visioniq-runtime` service account as a direct `roles/secretmanager.secretAccessor` principal on each application DB password secret using `gcloud secrets add-iam-policy-binding`. Each command returned `Updated IAM policy` and listed both migration and runtime identities. No grant to the administrator-password secret and no project-wide grant evidenced. Recorded audit commands, verification, least-privilege trade-off, pending effective-access test and authorized rollback considerations.

| 30 Oct 2026 | **1.12** | **Pipeline source prepared (no cloud-resource change):** Public FND-04 feature branch and draft PR #2 now contain separate Cloud Build image-builder, approval-gated application-deployer, and setup/runbook. Recorded intended usage of existing Artifact Registry, `vpc-con`, runtime identity and database secrets; preserved blockers for corporate GitHub connection, IAM approval, runtime

identity decision and authenticated application/database validation. No triggers, images or Cloud Run application services claimed deployed. |

| 30 Oct 2026 | **1.13** | **Cloud Build connection failure and deferral captured:** Recorded two actual failed `visioniq-build-dev` creation operations, code 9 / Secret Manager `secrets.create` denial, the Cloud Build P4SA's five direct roles, Analyzer grant vs Troubleshooter UNKNOWN (conditional allow; invisible PAB), and the operator-confirmed UI creation/grant of `visioniq-build-token-dev`. Captured corporate GitHub App installation restriction, lack of trigger/deployment, PAT cleanup review, and explicit decision to keep **all FND-04 user stories OPEN**. No successful application deployment implied. |

Maintenance after every material Google Cloud change: update the resource register, date, actual effective configuration, rationale, exact command/Console path, outcome and evidence, permissions granted/requested, rollback/recovery considerations, and changed document revision. Do not overwrite previously verified evidence with guessed status. Store screenshots and sanitized logs in the approved corporate evidence location; do not embed sensitive tokens or individually identifiable metadata.

Source-of-truth references: `db/README.md`, `db/manage.py`, `db/cloud_manage.py`, `db/migrations/env.py`, `db/Dockerfile`, `docker-compose.yml`, `.env.example` and prior Console/CLI output from 29–30 October 2026. This document records observed changes; it does not query live Google Cloud state by itself.

Date	Document update	Reason
30 Oct 2026	Reorganized as permanent `VisionIQ_Cloud_Structure` with a navigable service-based contents page	Avoid milestone-specific master documents; preserve detailed resource configuration, screenshots, commands and historical troubleshooting in one continuously updated cloud reference.

Appendix A. Reference Command Catalog

The commands below preserve the prior detailed cloud inventory and deployment-reference material. Their usage categories matter: **read-only inventory** can collect configuration evidence; **build, push, update, execute and IAM writes** modify real resources and require approval.

The following commands are a **reproducible inspection/runbook catalog**, because the complete 29-30 October terminal history is not available. The resource names come from the previous records. **No mutating command below was independently confirmed as the exact one typed during FND-03.** Use the read-only inventory commands in Section 13 first and request approved IAM changes when required.

Step A - select and verify the correct project (read-only)

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION  = 'europe-west3'
gcloud auth list
gcloud config list
gcloud projects describe $PROJECT --format='yaml(projectId,name,projectNumber)'
```

The earlier laptop initially had a different active project. An unsuccessful `gcloud config set project` was a **project access** problem, not an instruction to create another project. For approved access, select the corporate account and explicitly pass `--project` on all inspection commands.

Step B - Google Cloud APIs / resource list (read-only inventory)

```
gcloud services list --enabled --project=$PROJECT --format='table(config.name)'
gcloud artifacts repositories list --project=$PROJECT --location=$REGION
gcloud iam service-accounts list --project=$PROJECT --format='table(email,displayName,disabled)'
gcloud secrets list --project=$PROJECT --format='table(name,createTime)'
gcloud run jobs list --project=$PROJECT --region=$REGION
```

Record exact enabled API names from this output. Do not assert all optional APIs were created during FND-03; some may predate VisionIQ and can be shared by other teams.

Step C - Cloud SQL, database, user and backup evidence (read-only)

```
gcloud sql instances describe visioniqdev --project=$PROJECT   --
format='yaml(name,state,region,databaseVersion,connectionName,settings.tier,ipAddresses)'
gcloud sql databases list --instance=visioniqdev --project=$PROJECT
gcloud sql users list --instance=visioniqdev --project=$PROJECT
gcloud sql backups list --instance=visioniqdev --project=$PROJECT
```

The instance was created **manually through Console**; a `gcloud sql instances create` command is **not** part of the verified history. If recreating from scratch in a new, approved project, choose machine, region, edition, maintenance, private IP and backup settings through the organization’s provisioning process. Never recreate the existing instance just to obtain screenshots.

Step D - existing private network and route (read-only)

```
$NETWORK = 'vpc-mts-wgh-hkt-mtswghhkt-dev'
$SUBNET  = 'subnet-mts-wgh-hkt-mtswghhkt-dev'
gcloud compute networks describe $NETWORK --project=$PROJECT
gcloud compute networks subnets describe $SUBNET --region=$REGION --project=$PROJECT
gcloud compute networks peerings list --network=$NETWORK --project=$PROJECT
gcloud compute addresses list --global --project=$PROJECT   --filter='purpose=VPC_PEERING' --
format='table(name,address,prefixLength,network)'
gcloud compute firewall-rules list --project=$PROJECT   --filter='network:vpc-mts-wgh-hkt-mtswghhkt-dev AND
direction:EGRESS'
```

Historical access failure: The Service Networking peering API returned a `servicenetworking.services.get` permission denial, while Compute Engine peerings listing was usable. Do **not** interpret this as peering being absent or solve it by creating a new VPC. Request viewing rights when necessary. Record the service-networking and shared-service peering states, observed as ACTIVE.

Step E - distinguish connector from Direct VPC (read-only)

```
# Earlier connector inventory (may be absent or have limited visibility):
gcloud compute networks vpc-access connectors list --region=$REGION --project=$PROJECT
# The connector existed in earlier work. Only run describe if it remains present:
gcloud compute networks vpc-access connectors describe vpc-con --region=$REGION --project=$PROJECT

# Decisive evidence: actual job networking configuration:
gcloud run jobs describe visioniq-db-migrate --region=$REGION --project=$PROJECT --format=yaml
```

The **saved job YAML (generation 10)** includes `run.googleapis.com/vpc-access-connector` pointing to `vpc-con`. The **successful execution's own Networking tab separately confirms the connector**, the VPC, its `-connector` subnet and private-IP-only routing. Direct VPC egress is **not** the method used by the successful migration. This is a verified finding, not an inference from the job's later configuration or from TCP success alone.

Step F - Artifact Registry image inventory (read-only)

```
gcloud artifacts docker images list   europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev   --
project=$PROJECT --include-tags
```

Recorded successful image tag:

```
europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev/visioniq-intelligence:fnd03-netretry
```

Historical limitation: Successful job used that image. Later code added `db/Dockerfile` for a dedicated migration image. The standalone image was **not confirmed deployed**; the next change must record an exact image digest, build provenance and job-update event.

Step G - reference build / deploy / execute workflow (CHANGES CLOUD STATE)

> **Do not execute the block below during documentation or daily development.** It illustrates the sequence and flags that must be captured for an *approved future deployment*, not exact historical commands. Use unique immutable tags, approved build method, IAM and change ticket.

```
# REFERENCE ONLY - image creation and job update modify shared resources.
$PROJECT='gbr-mts-wgh-hkt-dev'
$REGION='europe-west3'
$IMAGE="${REGION}-docker.pkg.dev/${PROJECT}/team-visioniq-docker-dev/visioniq-db-migrate:APPROVED_TAG"

# Example from repository root only when Dockerfile exists and source has been reviewed:
docker build -f db/Dockerfile -t $IMAGE .
gcloud auth configure-docker "${REGION}-docker.pkg.dev"
docker push $IMAGE

# Example only: update only the image, preserving the existing approved `vpc-con` connection.
gcloud run jobs update visioniq-db-migrate --project=$PROJECT --region=$REGION --image=$IMAGE

# Example only - affects database state. Need reviewed migration/change approval:
gcloud run jobs execute visioniq-db-migrate --project=$PROJECT --region=$REGION --wait
```

PowerShell note: The reference invocations are deliberately shown on single lines, to remain copy-paste compatible in PowerShell. Do not run state-changing reference commands until reviewed and approved.

Step H - migration image and job configuration to record

Item	Effective value in 30 Oct historical record	Evidence requested
Job name	`visioniq-db-migrate`	Job details screenshot + read-only describe
Migration service account	`visioniq-db-migrate@...`	Job Security tab (no key)
Image	`visioniq-intelligence:fnd03-netretry`	Job container image digest
Command	`python -m db.cloud_manage all --instance ...`	Job container settings
Cloud SQL database	`visioniq_dev`	SQL Studio query

Item	Effective value in 30 Oct historical record	Evidence requested
DB host / port	`10.102.16.8:5432`	Sanitized logs
Database TLS	`sslmode=require`	Config without passwords
Secret names	`visioniq-db-admin-password`, `visioniq-db-api-password`, `visioniq-db-intelligence-password`	Secret Manager names + job mappings
Parallelism / tasks	1 / 1	Cloud Run job settings
Timeout / retries	10 min / 0	Cloud Run job settings
Network route	**Successful execution VERIFIED `vpc-con` and private-IP-only routing**, matching current saved job YAML	Historical change sequence for failed attempts remains to be recovered

> **G-04 Artifact Registry:** repository list and final image tag/digest (hide unrelated repositories as needed).

G-05 UPDATE: Container command, image, task count and CPU/memory captured in Section 8; execution status and network method confirmed separately. **Immutable image digest** remains pending.

G-06 UPDATE: Three secret references and migration service-account friendly label captured in Section 8. Full identity, IAM policy and `visioniq-runtime` account remain to be evidenced with safe metadata only.

Appendix B. Technical Glossary

Term	Meaning in VisionIQ
Project	Google Cloud administrative and billing boundary (`gbr-mts-wgh-hkt-dev`).
Region	Geographic location of resources (`europe-west3`).
VPC	Private software-defined network containing authorized routes/subnets.
Subnet	Regional IP-address segment used for Cloud Run Direct VPC egress.
PSA	Google-managed private service connection behind Cloud SQL private IP.
Connector	Serverless VPC Access component; separate option from Direct VPC egress.
Cloud Run Job	Scheduled/on-demand execution that terminates when work ends; not an HTTP API.
Cloud Run Service	Long-running request-serving application deployed as revision(s); FND-04 scope.
Artifact Registry	Repository for container images; image tags may change, digests are immutable.

Term	Meaning in VisionIQ
IAM service account	Cloud identity with Google API permissions; not a PostgreSQL user.
Secret Manager	Storage/reference for sensitive DB credentials.
Alembic	Python migration versioning; `public.alembic_version` shows applied revision.
pgvector	PostgreSQL extension for vector data, created/verified during FND-03.
FTS	PostgreSQL full-text search functionality; separate from embedding search.

Appendix C. Outstanding Documentation Evidence

- **FND-04 is DEFERRED / all user stories OPEN:** Two failed `visioniq-build-dev` creation operation IDs and the Cloud Build P4SA role list are now recorded; preserve the original, sanitized Cloud Audit Log denial and full Policy Troubleshooter expanded policy details when authorized. Do not record token values.
- **Cloud Build GitHub token secret:** User-confirmed `visioniq-build-token-dev` secret and P4SA accessor grant; record actual secret version metadata and approved revocation/deletion of its PAT if the GitHub App connection is abandoned, without accessing the payload.
- **Missing corporate GitHub authorization:** No Google Cloud Build GitHub App installation ID, connection, trigger, source build or VisionIQ Cloud Run application service is verified. Deployment requires approved source/build route and separately provisioned build/deploy identities.
- Copy sanitized original Cloud Run failure and success logs, with timestamps and execution IDs.
- **Successful migration networking VERIFIED**: execution's Networking screenshot shows `vpc-con`, connector subnet and private-only egress; saved generation 10 job YAML matches. **Historical troubleshooting sequence** around earlier Direct VPC attempts remains open pending failed-run logs/Audit records.
- Confirm VPC shared-network authorization and the final firewall-rule intent with corporate cloud administrators.
- Capture exact Docker image digest and final migration CLI flags from deployed job.
- **Application-secret direct IAM policies UPDATED on 30 Oct:** `visioniq-db-api-password` and `visioniq-db-intelligence-password` now each list both `visioniq-db-migrate` and `visioniq-runtime` under `roles/secretmanager.secretAccessor`. Updated policies are confirmed from CLI command responses; verify effective runtime access at deployment, and check admin-secret metadata only if authorized. Never collect secret contents.
- Fresh-volume migration/rerun acceptance was **VERIFIED in the public CI isolated PostgreSQL job** on 30 Oct 2026 (GitHub Actions run `41021168409`, all four jobs passed). Preserve logs as evidence; the same test has not been independently repeated on the corporate runner.
- Corporate PR completion was user-confirmed on 30 Oct 2026; capture the corporate merge SHA and approved alternative CI evidence when available.
- Update future application service deployments under Section 9 in this same master guide, without rewriting prior history.

VisionIQ — Google Cloud Structure

FND-04 Deployment Update & Repeatable Cloud Operations Runbook

Update date: 31 October 2026
Companion to: `VisionIQ_Cloud_Structure.md` (30 October 2026 master snapshot)
Project: `gbr-mts-wgh-hkt-dev`
Region: `europe-west3` (Frankfurt)
Scope: FND-04 manual release, evidence, troubleshooting, verification, reusable Google Cloud commands, and outstanding acceptance.
Source/evidence: Deployment logs and screenshots supplied during the 31 October 2026 deployment. Historical FND-03 configuration is taken from the prior master guide. This is not a live cloud inventory.

> **Document control:** This update supersedes the previous master guide's statements that `visioniq-api` and `visioniq-intelligence` are *not deployed*. It does **not** supersede the detailed FND-03 sections or imply that runtime HTTP/database acceptance has passed. Integrate this update into the corresponding service chapters and appendices when republishing the master guide. Do not delete the older incident and provisioning history.

Contents

1. Executive status and acceptance boundary
2. Updated architecture and resource inventory
3. FND-04 release record and deployment evidence
4. Exact manual release procedure — repeatable from Windows PowerShell
5. Cloud Run service configuration and operations
6. IAM, Secret Manager, private network and Cloud SQL
7. Actual troubleshooting history and resolution
8. Read-only Google Cloud command catalog
9. Cloud SQL and security acceptance procedures — pending
10. Health/readiness testing options — pending
11. Rollback and recovery
12. Evidence register, operational change log and next actions

---

1. Executive status and acceptance boundary

FND-04 manual deployment: DEPLOYED and metadata-verified. The agreed separate automation story **FND-AUTO** remains OPEN/BLOCKED due to corporate Cloud Build / Developer Connect permissions. The Cloud Run deployment has **not** been end-to-end accepted. The decision to defer endpoint and database testing must remain visible and tracked.

Work item	Status at 31 Oct 2026	Actual evidence
Local Docker build (3 images)	Complete	Docker local build output

Work item	Status at 31 Oct 2026	Actual evidence
Upload of images to Artifact Registry	Complete	`gcloud artifacts docker images list --include-tags` output at 18:54–18:55 IST
Cloud Run deployment: .NET API	Complete	Revision `visioniq-api-00001-9gm`, 100% traffic
Cloud Run deployment: Python Intelligence	Complete	Revision `visioniq-intelligence-00001-zfn`, 100% traffic
Revision health, ingress and IAM metadata	Complete	`03-verify-deployment.ps1` output
.NET 10 and Python process startup	Complete	Console application logs + successful TCP startup probe
Authorized HTTP `/health` (both)	**Deferred; not passed**	Calls from laptop through local gcloud proxy returned Google Frontend HTTP 404
Authorized HTTP `/ready` with real DB check (both)	**Deferred; not passed**	No completed authorized in-network response
Real application Cloud SQL reads / role isolation	**Deferred; not passed**	No successful queries from application identities captured
Cross-schema/secret isolation acceptance	**Deferred; not passed**	No runtime denial tests captured
Corporate shared service-account / secret IAM security approval	**Approval still to be recorded**	Provisioning and successful deployment are not formal approval
FND-04 guide and rollback notes	Included here	This document

Closeout policy: FND-04 may be marked complete *only as the manual deployment capability* if the remaining HTTP, database, and security acceptance are explicitly moved into separately tracked work with stated prerequisites. Never mark these tests passed. Do not use this release as proof of LRN-01–07 cloud readiness.

2. Updated architecture and resource inventory

```
Git main (reviewed commit / immutable git SHA)
   |
   | manual local Docker builds / approved operator
   v
Artifact Registry: team-visioniq-docker-dev (europe-west3)
   |-- visioniq-api:<git-sha>
   |-- visioniq-intelligence:<git-sha>
   `-- visioniq-db-migrate:<git-sha>   [uploaded; NOT run during FND-04]
        |
        +--> Cloud Run Service: visioniq-api
        |      Revision: visioniq-api-00001-9gm
        |      Container port 5100; internal ingress; authenticated
        |      Runtime identity: visioniq-runtime
        |      DB role intended: visioniq_api
        |
        +--> Cloud Run Service: visionIQ-intelligence (actual resource: visioniq-intelligence)
        |      Revision: visioniq-intelligence-00001-zfn
        |      Container port 8000; internal ingress; authenticated
        |      Runtime identity: visioniq-runtime
        |      DB role intended: visioniq_intelligence
        |
        `--> Pre-existing Cloud Run JOB: visioniq-db-migrate
               Separate migration identity; FND-04 made NO job update or execution
```

```
Both app services ---> Serverless VPC Access connector vpc-con [READY]
                       ---> corporate VPC / private routing
                       ---> Cloud SQL visioniqdev:5432 [10.102.16.8]
                       ---> database visioniq_dev

Runtime secrets: visioniq-db-api-password:2 and
                 visioniq-db-intelligence-password:2
                 [numeric versions enabled at release; version 1 disabled]

Separate future: GitHub/Developer Connect --> Cloud Build --> automated deploy
                 [FND-AUTO; NOT implemented or required for FND-04 manual deploy]
```

> Spelling note: the actual Google Cloud names are **`visioniq-api`** and **`visioniq-intelligence`**. The source services are .NET API and Python Intelligence, respectively.

Resource snapshot

Resource	Verified / recorded value	Notes
Google Cloud project	`gbr-mts-wgh-hkt-dev`	Shared corporate development environment; unrelated resources must remain untouched
Region	`europe-west3`	Frankfurt
Artifact Registry repository	`team-visioniq-docker-dev`	ONE repository holds THREE separately named images
Cloud SQL	`visioniqdev`; database `visioniq_dev`	Earlier FND-03 provisioning; private IP `10.102.16.8`, port 5432
VPC connector	`vpc-con`	`READY` on 31 October
Runtime service account	`visioniq-runtime@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com`	`iam.serviceAccounts.actAs` grant needed for deployer; shared-secret approval still to be documented
Migration service account	`visioniq-db-migrate@...`	Exact IAM/job bindings documented in original FND-03 guide
Runtime password secrets	`visioniq-db-api-password` and `visioniq-db-intelligence-password`	Both **enabled version 2**; version 1 disabled at time of deployment
API Cloud Run	`visioniq-api`	New service; private, authenticated, port 5100
Intelligence Cloud Run	`visioniq-intelligence`	New service; private, authenticated, port 8000
Database migration Cloud Run Job	`visioniq-db-migrate`	Pre-existing; FND-04 built/pushed migration image but did not execute or update job

3. FND-04 release record and deployment evidence

Source branch observed at deployment: `FND-04-manual-deployment`
Approved/used image tag and local Git HEAD: `02779fd00c0c0800443486a7263c7ee19d6f8fc0`
Earlier local build commit: `234c28dcd734d3dec4b5907b07861d358d282bbb` (different earlier checkout; **not** the uploaded release tag).

Uploaded images — tag-to-digest record

The multi-row `docker images list --include-tags` output also contains index/config/manifest objects with empty tags; these are normal OCI artifacts. The release-identifying rows are the **tagged** entries:

Image	Tag	Release manifest/index digest shown
`visioniq-api`	`02779fd00c0c0800443486a7263c7ee19d6f8fc0`	`sha256:d5cde7a16243f34b4b3bbdf563a38875ab44001c7ad903035154f6f842086f6a`
`visionIQ-intelligence` (registry name `visioniq-intelligence`)	Same	`sha256:9460a72f6d3f18446929cf23050e8beb3d5432268988fbcc7db5e007a24a4384`
`visioniq-db-migrate`	Same	`sha256:b2c2cbcd1e1f29fee960b8de10ae6cc73a850241add979bdaaee028f133a3b43`

Important: A tagged registry digest may identify an OCI image index instead of a single platform manifest. Confirm the selected deployment digest with `gcloud run revisions describe` before using an immutable digest in a rollback plan; preserve the exact artifact reference that Cloud Run reports.

Cloud Run deployment record

Service	Revision	Status	Traffic	URL recorded by Cloud Run
`visioniq-api`	`visioniq-api-00001-9gm`	READY	100%	`https://visioniq-api-494928274920.europe-west3.run.app`
`visioniq-intelligence`	`visioniq-intelligence-00001-zfn`	READY	100%	`https://visioniq-intelligence-494928274920.europe-west3.run.app`

The `gcloud run services proxy` command resolved corresponding alternate `a.run.app` URLs:

- `https://visioniq-api-mtoeepkfva-ey.a.run.app`
- `https://visioniq-intelligence-mtoeepkfva-ey.a.run.app`

Use `gcloud run services describe ... --format="value(status.url)"` to obtain the current canonical URL; **do not hard-code URL variants as interchangeable authentication audiences without checking**.

Start-up observation (not database acceptance)

- .NET: `Now listening on: http://0.0.0.0:5100`; `Application started`; Cloud Run TCP startup probe passed; warning about `HTTP_PORTS 8080` being overridden by URL binding to 5100 is **nonfatal**.
- Python: Uvicorn listening on `0.0.0.0:8000`, application startup complete, TCP startup probe passed.
- Earlier Cloud Audit Log `iam.serviceaccounts.actAs` permission-denied entries relate to failed deployment before IAM grant; later successful revisions establish deployment was recovered.
- Neither startup probe establishes `/health`, `/ready`, SQL SELECT, database credential scope, or cross-schema security.

4. Exact manual release procedure — Windows PowerShell

Run from the **VisionIQ repository root**, with Docker Desktop/engine working and the corporate `gcloud` identity selected. These commands are preserved for reproducibility. Review scripts in the repository before running; deployed code and infrastructure may have evolved since this snapshot.

4.1 Read-only prerequisites

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION  = 'europe-west3'
$REPO    = 'team-visioniq-docker-dev'

# Confirm branch/commit and local state
git branch --show-current
git rev-parse HEAD
git status --short
$SHA = (git rev-parse HEAD).Trim()

# Use approved corporate identity
gcloud auth list
gcloud config get-value project

# Docker engine / availability
docker version
docker info

# Registry and networking inventory
gcloud artifacts repositories describe $REPO --project=$PROJECT --location=$REGION --format='table(name,format)'
gcloud compute networks vpc-access connectors describe vpc-con --project=$PROJECT --region=$REGION --
format='value(state)'

# Review runtime credential versions; do not read secret payloads
gcloud secrets versions list visioniq-db-api-password --project=$PROJECT
gcloud secrets versions list visioniq-db-intelligence-password --project=$PROJECT
```

Safety: `-Push` intentionally refuses tracked local changes. Do not use `git reset --hard` or `git clean -fd` to silence the guard. Investigate `git status --short` and preserve work first.

4.2 Build images locally (no upload)

```
powershell -ExecutionPolicy Bypass -File `
  .\scripts\manual-deploy\01-build-push.ps1 -BuildOnly
```

This builds `visioniq-api`, `visioniq-intelligence`, and `visioniq-db-migrate` locally. It does **not** deploy or execute migrations.

4.3 Push all three images (writes to Artifact Registry)

```
$SHA = (git rev-parse HEAD).Trim()
powershell -ExecutionPolicy Bypass -File `
  .\scripts\manual-deploy\01-build-push.ps1 `
  -Push -ReviewedCommit $SHA
```

The script enforces clean tracked Git state and exact reviewed SHA. Push image tags are tied to the specific Git commit, not to mutable `latest`. Do not assume printed image names imply push success; look for successful upload and verify remotely.

4.4 Verify pushed tags and digests (read-only)

```
gcloud artifacts docker images list `
  europe-west3-docker.pkg.dev/gbr-mts-wgh-hkt-dev/team-visioniq-docker-dev `
  --project=$PROJECT --include-tags
```

Find three tagged images with the **same approved SHA**. Multiple untagged digest rows are normal.

4.5 Plan deployment (read-only script mode)

```
$SHA = '02779fd00c0c0800443486a7263c7ee19d6f8fc0'
# THESE VERSIONS ARE A DATED RELEASE RECORD. Recheck versions for a later release.
$API_SECRET_VERSION = 2
$INT_SECRET_VERSION = 2

powershell -ExecutionPolicy Bypass -File `
  .\scripts\manual-deploy\02-deploy-services.ps1 `
  -Commit $SHA `
  -ApiSecretVersion $API_SECRET_VERSION `
  -IntelligenceSecretVersion $INT_SECRET_VERSION
```

Expected plan excerpt: project, region, `visioniq-runtime`, private Cloud SQL `10.102.16.8:5432` via `vpc-con`, image tags, internal ingress, auth, service ports, `PLAN ONLY`.

4.6 Deploy application services (CLOUD WRITE; authorized change)

```
powershell -ExecutionPolicy Bypass -File `
  .\scripts\manual-deploy\02-deploy-services.ps1 `
  -Commit $SHA `
  -ApiSecretVersion $API_SECRET_VERSION `
  -IntelligenceSecretVersion $INT_SECRET_VERSION `
  -Deploy
```

Actual 31 Oct result: Python Intelligence created first, then .NET API; both revisions READY, 100% traffic. **No migration Job execution.** Repeat only with reviewed image version, explicit secret versions, approved IAM and network configuration.

4.7 Metadata verification (read-only)

```
powershell -ExecutionPolicy Bypass -File `
  .\scripts\manual-deploy\03-verify-deployment.ps1 `
  -Commit $SHA
```

Verified 31 Oct: both revisions READY, expected image tags, ingress `internal`, no public Invoker IAM bindings. **This script does not certify real SQL role access or in-network HTTP.**

5. Cloud Run Services — deployed

5.1 Detailed service description and revision inventory

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'; $REGION = 'europe-west3'

gcloud run services list --project=$PROJECT --region=$REGION

gcloud run services describe visioniq-api `
  --project=$PROJECT --region=$REGION `
  --
format='yaml(metadata.name,status.url,status.latestReadyRevisionName,status.traffic,spec.template.spec.serviceAccountN
ame,spec.template.spec.containers)'

gcloud run services describe visioniq-intelligence `
  --project=$PROJECT --region=$REGION `
  --
format='yaml(metadata.name,status.url,status.latestReadyRevisionName,status.traffic,spec.template.spec.serviceAccountN
ame,spec.template.spec.containers)'

gcloud run revisions list --service=visioniq-api --project=$PROJECT --region=$REGION
```

```
gcloud run revisions list --service=visioniq-intelligence --project=$PROJECT --region=$REGION
```

The `visionIQ-api` typo above is included solely as an operator warning; use the lowercase corrected command. Preserve a service YAML snapshot before future deploys (avoid sharing env/secret data).

5.2 Runtime logs and observability

```
gcloud run services logs read visioniq-api --project=$PROJECT --region=$REGION --limit=100
gcloud run services logs read visioniq-intelligence --project=$PROJECT --region=$REGION --limit=100

gcloud run services describe visioniq-api --project=$PROJECT --region=$REGION `
  --format='yaml(status.conditions,status.latestReadyRevisionName)'
gcloud run services describe visioniq-intelligence --project=$PROJECT --region=$REGION `
  --format='yaml(status.conditions,status.latestReadyRevisionName)'
```

Console paths: Cloud Run → Services → `visioniq-api` or `visioniq-intelligence` → Observability → Logs; Revision History; Containers; Networking; Security. Use time-scoped logs to distinguish historical failed operations from a current deployment issue.

5.3 Internal ingress and authentication

The deployed services were explicitly verified as **internal ingress**, authenticated, with no public (`allUsers` / `allAuthenticatedUsers`) Invoker binding at service level. **Do not change to allow-all ingress or unauthenticated access merely to test with Postman.**

An empty service-level IAM policy response such as `etag: ACAB` means no direct binding was displayed; inherited project-level role grants can still be effective. During 31 October inspection, project-level `roles/run.invoker` roles were visible for four unrelated/shared service accounts; these are **not** blanket authorization to reuse their credentials for VisionIQ testing. A candidate identity needs approval and an allowed internal network path.

6. IAM, Secrets, VPC and Cloud SQL

6.1 IAM deployment failure and fix

Observed error: `Permission 'iam.serviceaccounts.actAs' denied on service account visioniq-runtime@...` during first `visioniq-intelligence` deployment. The deployer needed **Service Account User (`roles/iam.serviceAccountUser`) on the runtime SA**. After an IAM correction, deploying the same image tag succeeded. Service-account attachment permission is different from the runtime identity's ability to read its allowed secrets.

Diagnostic commands (read-only):

```
gcloud auth list
gcloud iam service-accounts describe `
  visioniq-runtime@$PROJECT.iam.gserviceaccount.com --project=$PROJECT

gcloud iam service-accounts get-iam-policy `
  visioniq-runtime@$PROJECT.iam.gserviceaccount.com --project=$PROJECT

gcloud run services get-iam-policy visioniq-api --project=$PROJECT --region=$REGION
gcloud run services get-iam-policy visioniq-intelligence --project=$PROJECT --region=$REGION

gcloud projects get-iam-policy $PROJECT `
  --flatten='bindings[].members' `
  --filter='bindings.role:roles/run.invoker' `
  --format='table(bindings.role,bindings.members)'
```

Possible remediation, ONLY by authorized IAM administration (changes permissions):

```
# A privileged administrator should assess/approve the least-privilege grant.
gcloud iam service-accounts add-iam-policy-binding `
  visioniq-runtime@gbr-mts-wgh-hkt-dev.iam.gserviceaccount.com `
  --project=gbr-mts-wgh-hkt-dev `
  --member='user:<approved-corporate-deployer-email>' `
  --role='roles/iam.serviceAccountUser'
```

Do **not** run the placeholder remediation verbatim. Never grant the role project-wide merely as a shortcut. Do not record personal/corporate email addresses in a public runbook.

6.2 Secret Manager

```
gcloud secrets list --project=$PROJECT --filter='name:visioniq'
gcloud secrets versions list visioniq-db-api-password --project=$PROJECT
gcloud secrets versions list visioniq-db-intelligence-password --project=$PROJECT

gcloud secrets get-iam-policy visioniq-db-api-password --project=$PROJECT
gcloud secrets get-iam-policy visioniq-db-intelligence-password --project=$PROJECT
```

Verified on 31 October: enabled numeric version **2** for both passwords; version **1** disabled. On later releases, re-read versions rather than hard-coding 2. The Cloud Run application identity had prior direct accessor grants on both application secrets; the corporate approval for this **shared-secret boundary** was not independently documented. **Do not run `gcloud secrets versions access` or copy secret payloads to evidence logs.**

6.3 Private network and database inventory

```
gcloud compute networks vpc-access connectors describe vpc-con `
  --project=$PROJECT --region=$REGION

gcloud sql instances describe visioniqdev --project=$PROJECT `
  --format='yaml(name,state,region,connectionName,databaseVersion,ipAddresses)'

gcloud sql databases list --instance=visioniqdev --project=$PROJECT

gcloud run jobs describe visioniq-db-migrate --project=$PROJECT --region=$REGION
```

This verifies configuration, **not** runtime SQL reachability. `vpc-con` `READY` and Cloud Run `READY` are insufficient alone to prove successful application database authentication.

7. Troubleshooting history (observed, not hypothetical)

Date / stage	Symptom	Root cause / interpretation	Action taken / future rule
Initial FND-04 push attempt	`Refusing to push images from a checkout with tracked local changes`	Script's intentional release safety guard	Check `git status --short`; preserve changes; only push a clean reviewed commit. Do not use destructive reset. Later push succeeded.
Early local build	Images printed with tag `234c28...` but no upload	`-BuildOnly` created local images	Upload performed later using reviewed `02779f...` SHA; confirm with registry list
First Cloud Run deploy	`iam.serviceaccounts.actAs` denied on `visionIQ-runtime` SA	Deployer could not attach approved runtime service identity	IAM corrected via approved route; retry succeeded; record grant approval

Date / stage	Symptom	Root cause / interpretation	Action taken / future rule
Cloud Run startup log	.NET warning about `HTTP_PORTS=8080` overridden by `http://0.0.0.0:5100`	App explicitly listens on correct container port	Non-fatal; TCP startup succeeded. Confirm desired server binding; don't change just for warning
Python startup	Uvicorn on `0.0.0.0:8000`	Intended port and process	Startup passed; DB not yet checked
Postman via local gcloud proxy	Both `/health` returned `HTTP/1.1 404 Not Found`, `Server: Google Frontend`	Consistent with internal ingress from an unapproved laptop network; request not proven to reach app	Do not interpret as app-route 404; do not make services public. Use approved internal caller with auth.
IAM inspection	Service-level `get-iam-policy` only `etag: ACAB`	No direct bindings shown; inheritance still possible	Inspect project-level bindings; do not infer invoker access from empty service-level policy
Project IAM inspection	Four existing project-level Invoker service accounts listed	Shared corporate identities, not VisionIQ test authorization	Do not impersonate or reuse without explicit approval
Earlier FND-03 Cloud Build integration	Developer Connect/Cloud Build connection creation denied around service-agent Secret Manager permissions	Corporate authorization blocker	FND-AUTO remains separate; manual deploy path is the FND-04 solution

Troubleshooting command quick path: `git status --short` → `gcloud auth list` → `gcloud config get-value project` → registry image tag listing → `gcloud run services describe` → `get-iam-policy` → `logs read` → `vpc-con describe` → secret version listing. Stop when evidence identifies a missing corporate permission; do not work around controls with broad grants.

8. Repeatable gcloud command catalog — read-only first

This catalog is intended for a future operator who must re-check everything without remembering what was typed during the hackathon.

8.1 Daily / release inventory (safe read-only)

```
$PROJECT = 'gbr-mts-wgh-hkt-dev'
$REGION = 'europe-west3'
$REGISTRY = "${REGION}-docker.pkg.dev/$PROJECT/team-visioniq-docker-dev"

gcloud auth list
gcloud config get-value project
gcloud config list
gcloud projects describe $PROJECT
gcloud run services list --project=$PROJECT --region=$REGION
gcloud run jobs list --project=$PROJECT --region=$REGION
gcloud artifacts repositories list --project=$PROJECT --location=$REGION
gcloud artifacts repositories describe team-visioniq-docker-dev --project=$PROJECT --location=$REGION
gcloud artifacts docker images list $REGISTRY --project=$PROJECT --include-tags
gcloud secrets list --project=$PROJECT --filter='name:visioniq'
gcloud iam service-accounts list --project=$PROJECT --filter='email:visioniq'
gcloud compute networks vpc-access connectors describe vpc-con --project=$PROJECT --region=$REGION --
format='value(state)'
gcloud sql instances list --project=$PROJECT
gcloud sql instances describe visioniqdev --project=$PROJECT
```

8.2 Specific service evidence (safe read-only)

```
gcloud run services describe visioniq-api --project=$PROJECT --region=$REGION
gcloud run services describe visioniq-intelligence --project=$PROJECT --region=$REGION
gcloud run services get-iam-policy visioniq-api --project=$PROJECT --region=$REGION
gcloud run services get-iam-policy visioniq-intelligence --project=$PROJECT --region=$REGION
gcloud run services logs read visioniq-api --project=$PROJECT --region=$REGION --limit=100
gcloud run services logs read visioniq-intelligence --project=$PROJECT --region=$REGION --limit=100
gcloud run revisions list --service=visioniq-api --project=$PROJECT --region=$REGION
gcloud run revisions list --service=visioniq-intelligence --project=$PROJECT --region=$REGION
```

8.3 CI and future automation diagnosis (safe read-only)

```
gcloud builds triggers list --project=$PROJECT --region=$REGION
gcloud builds connections list --project=$PROJECT --region=$REGION
gcloud services list --enabled --project=$PROJECT `
  --filter='config.name:(cloudbuild.googleapis.com OR developerconnect.googleapis.com OR run.googleapis.com OR
secretmanager.googleapis.com)'
```

Do not conclude global absence from regional zero results. Treat these as scoped snapshots, not proof that a service/trigger does not exist elsewhere.

8.4 Local proxy/Postman attempt (diagnostic, not a security bypass)

```
# Two terminal windows, approved operator identity:
gcloud run services proxy visioniq-api --project=$PROJECT --region=$REGION --port=8510
gcloud run services proxy visioniq-intelligence --project=$PROJECT --region=$REGION --port=8800

# Third terminal; only if both proxies are running:
curl.exe -i --max-time 20 http://127.0.0.1:8510/health
curl.exe -i --max-time 20 http://127.0.0.1:8800/health
```

The actual result on 31 Oct was **Google Frontend HTTP 404 for both calls**, so this is **not a validated way to test these internal-ingress services from the current laptop**. Use `Ctrl+C` to stop proxies. Postman can use `http://127.0.0.1:8510/health` and `:8800/health` only when the proxy also has an approved internal network route and the caller is authorized. Do not mistake a running proxy for successful HTTP reachability.

8.5 Command safety legend

- **Read-only:** `describe`, `list`, `get-iam-policy`, `logs read`, registry images list, local `git status`.
- **Cloud writes:** `gcloud run deploy`, `gcloud run services update-traffic`, `gcloud run services add-iam-policy-binding`, `gcloud iam service-accounts add-iam-policy-binding`, `gcloud secrets ... add-iam-policy-binding`, `gcloud services enable`, `gcloud workflows deploy`, `gcloud run jobs execute`.
- **Potentially destructive:** `gcloud run services delete`, `gcloud run jobs delete`, `gcloud sql instances delete`, secret versions disable/destroy, broad IAM removals. Never execute these from this guide without approved change planning.

9. Deferred Cloud SQL, database ACL and secret-isolation acceptance

These tests are **unexecuted** as of 31 October. They should be captured in a follow-up validation story with explicit scope and evidence; they should run **before** live cloud demos or reliance on Cloud SQL.

Test	Execution condition	Pass criterion	Evidence required
API readiness	Authorized in-network caller using valid Cloud Run OIDC identity	`/ready` HTTP 200 *and* readiness code really checks DB	HTTP status, sanitized response and DB log correlation
Intelligence readiness	Same	`/ready` HTTP 200 and DB check	Same
API database role	Connect through approved private path with `visioniq_api` credentials	`SELECT current_user` and permitted SELECT succeed	Query output without passwords
Intelligence database role	Same with `visioniq_intelligence`	Authorized read succeeds	Query output
Cross-schema denial	Use schema/table names from actual grants; test SELECT/ WRITE privileges safely	Unauthorized permissions absent or denied	`has_table_privilege`/ `has_schema_privilege` and denied access
Secret isolation	IAM policy review plus approved negative access tests	Caller/test identity cannot read application DB passwords; each role respects documented policy	Redacted IAM evidence and denial

Baseline SQL for each approved role, after connecting over approved private network:

```
SELECT current_user, current_database();
SELECT 1 AS connection_test;
SELECT nspname AS schema_name,
       has_schema_privilege(current_user, oid, 'USAGE') AS schema_usage,
       has_schema_privilege(current_user, oid, 'CREATE') AS schema_create
FROM pg_namespace
WHERE nspname NOT LIKE 'pg_%'
  AND nspname <> 'information_schema'
ORDER BY nspname;
```

After discovering the actual schema/table set, use `has_table_privilege(current_user, 'schema.table', 'SELECT')` and approved read-only queries. A `SELECT 1` proves a DB session, **not** authorization against real app tables. A `/health` response proves liveness only if implemented as such. Avoid destructive tests and do not log credentials.

10. Deferred HTTP tests — decision and safe approaches

No test service account, Workflow or test Job was created in this FND-04 session. Creation was discussed but explicitly deferred to avoid unnecessary corporate resources. The existing corporate project has unrelated project-level Invoker service accounts; they must not be appropriated without approval.

Preferred order:

1. **Existing approved in-project/internal caller** with both `roles/run.invoker` permission and an allowed network origin. Authenticate using OIDC ID token with the current exact service URL as audience.
2. **Temporary Google Cloud Workflow** attached to a **pre-approved** service account, if no suitable existing caller is available; avoid retaining test-only resources after acceptance.
3. **Dedicated temporary test identity** only if security policy requires it and approved IAM can be granted.

Do **not** remove ingress restrictions or enable unauthenticated access simply to make Postman work. A Google Frontend 404 from an external workstation is not evidence of application endpoint behavior.

HTTP acceptance record template

Date/UTC time	Service	URL path	Caller identity (approved identifier)	Network origin	HTTP status	Body check	Result
Pending	visioniq-api	/health	Pending	Pending	Pending	Pending	Not tested
Pending	visioniq-api	/ready	Pending	Pending	Pending	Pending	Not tested
Pending	visioniq-intelligence	/health	Pending	Pending	Pending	Pending	Not tested
Pending	visioniq-intelligence	/ready	Pending	Pending	Pending	Pending	Not tested

11. Rollback and recovery

Current baseline: first deployed production-style revisions in this project (`visioniq-api-00001-9gm`, `visioniq-intelligence-00001-zfn`). There is **no earlier successful application revision available for rollback** as of this snapshot. Never claim an `update-traffic` rollback is available until a preceding ready revision exists.

Inspect revision and traffic state first (read-only)

```
gcloud run revisions list --service=visioniq-api --project=$PROJECT --region=$REGION
gcloud run revisions list --service=visioniq-intelligence --project=$PROJECT --region=$REGION

gcloud run services describe visioniq-api --project=$PROJECT --region=$REGION `
  --format='yaml(status.latestReadyRevisionName,status.traffic)'
gcloud run services describe visioniq-intelligence --project=$PROJECT --region=$REGION `
  --format='yaml(status.latestReadyRevisionName,status.traffic)'
```

After future deployments, if a known-good earlier revision still exists, an authorized release operator can redirect traffic back after reviewing current and previous revisions, running health tests and recording the change. Syntax example only (DO NOT run unless earlier known-good revision exists):

```
# Illustrative, state-changing; verify exact revision and authorization first.
# gcloud run services update-traffic visioniq-api `
#   --to-revisions=<KNOWN_GOOD_REVISION>=100 --project=$PROJECT --region=$REGION
```

Alternative recovery requires redeploying a previously validated, retained image tag/digest using approved secret versions and reviewed scripts. **Database migrations cannot be reversed just by rolling back a Cloud Run service.** FND-04 itself did not run migrations. Keep migration and application release procedures separate. Record the former revision, traffic split, image digest, change approval and verification evidence for each release.

12. Evidence register, change log and next actions

Evidence register

ID	Captured evidence	Status	Recommended source / screenshot
E-FND04-01	Clean Git HEAD and source branch	Observed text output	Git command output, redacted where necessary

ID	Captured evidence	Status	Recommended source / screenshot
E-FND04-02	Local three-image build	Observed local console output	Docker script terminal capture
E-FND04-03	Registry tagged images/ digests	**Captured in conversation output**	Artifact Registry → Images + terminal listing
E-FND04-04	Cloud Run API ready revision	**Captured in deploy log**	`visioniq-api-00001-9gm` service Overview / Revision History
E-FND04-05	Cloud Run Intelligence ready revision	**Captured in deploy log**	`visioniq-intelligence-00001-zfn` service Overview / Revision History
E-FND04-06	API .NET startup and TCP probe	**Screenshot supplied**	Cloud Run API → Observability → Logs
E-FND04-07	Python startup and TCP probe	**Screenshot supplied**	Cloud Run Intelligence → Observability → Logs
E-FND04-08	Metadata security verification	**Captured in script output**	`03-verify-deployment.ps1`, ingress/IAM
E-FND04-09	Initial actAs denial / later recovery	**Captured in terminal/audit logs**	Cloud Run audit logs; redact identities
E-FND04-10	Laptop proxy HTTP 404	**Captured in terminal output**	`curl.exe -i` and Google Frontend response
E-FND04-11	Authorized in-network 4 HTTP endpoint tests	**PENDING**	Future test output
E-FND04-12	SQL role/isolation/secret access checks	**PENDING**	Approved SQL client/IAM evidence
E-FND04-13	Formal security approval for shared runtime secret access	**PENDING RECORD**	Corporate authorization reference
E-FND04-14	Image digest used by live revision (Cloud Run status)	**PENDING CAPTURE**	`gcloud run revisions describe ...`

Change log

Date	Change	Result
29–30 Oct 2026	FND-03 Cloud SQL/PSA/VPC/Secret Manager/service accounts/migration Job; Cloud Build integration attempts	Retained in older master guide; app services not deployed as of 30 Oct
31 Oct 2026	Git-backed manual deployment scripts, local Docker build and push to `team-visioniq-docker-dev`	Three tagged release images uploaded
31 Oct 2026	IAM deploy failure `iam.serviceaccounts.actAs`	Corrected through approved IAM path; retry successful
31 Oct 2026	Created `visioniq-intelligence` and `visioniq-api` Cloud Run services	First revisions READY, each 100% traffic
31 Oct 2026	Ran FND-04 metadata verification; inspected .NET/Python startup logs	Passed metadata and startup checks

Date	Change	Result
31 Oct 2026	Tried local gcloud proxies, Postman/curl cloud HTTP	Google Frontend 404 due likely internal network restriction; no application endpoint pass
31 Oct 2026	Decided to defer live HTTP and DB ACL tests	Manual deployment scope may close only with explicit tracked acceptance follow-up
Later	FND-AUTO corporate developer integration and cloud CI	Separately tracked; not part of manual deployment completion

Post-release review and next actions

1. Incorporate this chapter into `VisionIQ_Cloud_Structure.md` and replace prior chapter 9 statements that the application services are absent. Keep historic 30 October statements labeled as dated context, not as current status.
2. Capture live deployed **revision image digests** and resource/configuration screenshots. Avoid copying secrets and unredacted policy membership into public repositories.
3. Create/confirm a tracked story for deferred **FND-04 cloud-runtime and security acceptance**. Define authorized internal test path and pass evidence.
4. Obtain and log corporate security acceptance of `visioniq-runtime` shared access to the application DB secrets or revise to separate identities if policy requires.
5. Keep FND-AUTO separate (Cloud Build / Developer Connect); do not retroactively claim it implemented.
6. Verify existing migration Cloud Run Job still points to an intended, approved migration image before any future migration execution. Do not update or execute that job as a side effect of FND-04 release.

Preservation rule: Keep every successful and failed command that materially explains provisioning, deployment, troubleshooting, testing or rollback in the appropriate section, including what a command changes and what output to expect. Never represent an example command as executed evidence. Maintain the cloud guide as one cumulative infrastructure source of truth while keeping developer-machine setup and daily local commands in their separate guides.